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Day Month Year

Contents

1	Introduction	7
2	Gravitational Wave Interferometers	8
3	Light and Gaussian Modes	9
4	High Power Laser Prototype for LIGO	10
5	Balanced Homodyne Detection for LIGO A+	11
5.1	What is BHD	12
5.2	BHD versus DC readout for A+	17
5.3	Local Oscillator Path Length Stability Requirement for A+	22
5.3.1	Local Oscillator Path	24
5.3.2	The Linear Response Function Between Differential Strain and Power at the AS Port	27
5.3.3	Converting the Sensitivity Curve for A+ into a Longitudinal Motion Requirement	28
5.3.4	Summary of LO path length Stability	32
5.4	Conclusion	32

6	Design of the LIGO A+ HAM6 layout	37
6.1	The Constraints and Requirements for the Optical Layout Within HAM6	38
6.1.1	Constraints	38
6.1.2	Requirements	41
6.1.3	Summary	43
6.2	The Chosen HAM6 Optical Layout	44
6.3	Visitations of the Robustness of the A+ HAM6 Layout to Mode Mis- match	51
6.3.1	A Description of Mode Matching in Terms of HG Modes . . .	51
6.3.2	Calculating Mode Matching Using The Gaussian Beam Pa- rameter	52
6.3.3	WS Space Visualisations	53
6.3.4	Visualising how Active Mode Matching Optics Expand the Area of WS Space Representing Acceptable Input Modes . . .	55
6.4	Conclusion	60
7	The Consequence of Radius of Curvature Errors within the SRC on Modes at the SRM.	61
8	In Air BHD experiments	62
9	In-vacuum experiments	63
10	Relevant Semiconductor and Photodiode Theory	64
11	Characterisation of InGaAs Photodiodes	65

12	Characterisation of 2um Photodiodes	66
13	Characterisation of InSb Photodiode	67
14	Conclusion	68
A	FINESSE Simulations	69
B	Algorithm For Finding a Three Optic Solution with 45° Gouy Phase	
	Separation between two of the Optics	70
	B.0.1 Method	71
C	Mode Matching Between 2 Astigmatic Beams	76
	C.1 Derivation of Mode Matching TM00 Beams	76
	C.2 Effect of Error in the Radius of Curvature of a Mirror on the defocus of a Beam	79
D	Sensing and Control in the HAM6 Layout	81
	D.0.1 Longitudinal Control	81
	D.0.2 Alignment Control	83
E	Quantum Optics	85

Status of Thesis

Note: red = pre first edit, orange = pre second edit and beyond, green = done, blue = may not make it into the thesis, or it may be moved to the appendix.

- Chapter 1: Not started. Will write last/almost last
- Chapter 2: Some ideas for content, but no words written yet.
- Chapter 3: Approaching first draft. Lots of work is still required to make clear that terms are properly defined.
- Chapter 4: Vague idea for structure, but almost no words have been written.
- Chapter 5: Passed first drafting. Still to apply Ken and Bryan's edits (see email from 29/09/2020 'quick question' for notes).
- Chapter 6: Passed first drafting. Awaiting Ken and Bryan's comments.
- Chapter 7: Passed first drafting. Still to apply Ken and Bryan's edits.
- Chapter 8: Not written or properly planned
- Chapter 9: Not written or properly planned

- Chapter 10: Not written or properly planned. Might not make it past planning stage
- Chapter 11: Not written or properly planned. May be relegated to the appendix
- Chapter 12: Significant amounts of thought has already gone into this.
- Chapter 14: Not started. Will write last/almost last
- Appendix A: Passed first drafting.
- Appendix B: Mostly written.
- Appendix C: Mostly written, but needs writing in the notation of Chapter 3
- Appendix D: Mostly written
- Appendix E: Likely to be culled

Chapter 1

Introduction

Chapter 2

Gravitational Wave Interferometers

Chapter 3

Light and Gaussian Modes

Chapter 4

High Power Laser Prototype for LIGO

Chapter 5

Balanced Homodyne Detection for LIGO A+

A major change between aLIGO and LIGO A+ will be the replacement of DC readout with Balanced Homodyne Detection (BHD). BHD plays a key role in many experiments that utilise non-classical states of light (for a recent example of such an experiment relating to gravitational wave interferometry, see [1]). The transition from aLIGO to LIGO A+ involves the redesign of many aspects of the interferometer, and elements of this effort aim to exploit quantum effects in order to increase the SNR of the detector. Therefore, the implementation of BHD is an attractive prospect for those who want to harness the quantum nature of light to enhance the detector.

It was found that, aside from the potential quantum mechanical benefits, BHD offers the potential to increase the SNR by reducing Signal Recycling Cavity (SRC) length noise coupling to the differential arm motion (DARM) signal. Even if LIGO were to never operate in a novel, quantum-mechanically enhanced way, the reduction

of SRC length noise is enough motivation to implement BHD. The longitudinal motion requirement for the optics that deliver the signal and local oscillator (LO) beams to the combining beam splitter was found to be $\leq 5 \times 10^{-17} \text{m}/\sqrt{\text{Hz}}$ at 100 Hz in order to avoid increasing the noise floor of the detector. Therefore, triple suspensions will be used to hang these optics.

This Chapter begins by exploring the fundamentals of BHD. Section 5.2 contrasts BHD with DC readout within the specific setting of LIGO A+ and Section 5.3 derives the phase noise requirement for the LO beam for A+. Section 5.4 concludes this Chapter.

5.1 What is BHD

BHD is a method of measuring information encoded by a light field, and it forms a key part of many quantum optics experiments. By combining a signal beam with a much stronger LO beam, phase modulations in the signal beam become amplitude modulations in the combined LO-signal beam (e.g. [2]). BHD can be configured to measure any quadrature of the signal beam which is vital for inferring the quantum state of a system and is required for more complex topologies of interferometer [3, 4, 5, 6]. BHD has been implemented to detect 11.6 dB of shot noise suppression to below 1 Hz due to the squeezing of vacuum fluctuations. [7].

All readout schemes for gravitational wave detectors rely on a LO field of some kind. The first generation of detectors used heterodyne readout (e.g. [8]). Heterodyne readout involves a modulation-demodulation scheme whereby the signal is mixed with a RF LO to a frequency where technical noise is no longer a concern. It

is then demodulated at the same modulation frequency to recover the original signal. This method comes with the draw back that the quantum shot noise is enhanced by the modulation-demodulation process [9].

The LO in a BHD readout scheme is a light field with the same optical frequency as the signal beam. The LO beam has a higher power than the signal beam and acts as a stable reference frequency. Ideally the LO would have no phase noise, however in reality there are fluctuations in the phase between the LO and the signal beam. The interference of the LO and signal beam causes this phase noise to become an amplitude modulation in the combined LO-signal beam[10]. This phase noise must be kept below the size of the phase fluctuations in the signal beam that are to be measured, which in A+ are determined by the shot noise of the light within the arm cavities. Unlike Heterodyne readout, BHD does not cause a fundamental increase in the noise floor of the measurement.

BHD is implemented by injecting the LO and signal beams into different input ports of a balanced beam splitter and placing photodiodes in the remaining output ports. The information encoded in an arbitrary quadrature of the signal beam can be selected by correctly setting the relative phase between the LO and signal beams. The relative phase between the LO and signal beam is also known as the *homodyne angle*. Subtracting the two photo-currents results in a measurement which is proportional to the selected quadrature of the signal beam. A simple BHD layout is shown in Figure 5.1.

The following blue text, along with figure 5.1, is something I want to rewrite at some point or add some extra technical detail to make it less hand wavy. For now,

I've followed the reasoning set out in [10] since this passed peer review, but I'm hoping to understand BHD and quantum optics well enough that I can reformulate it. The thing which I find weird about Sebastian's paper is the behaviour seems to be obtained through operators rather than quantum states. Also the semi-classicalness of it (the $a = \alpha + \delta a$ step) is weird/risky and I can't see a good reason, apart from it makes the maths easy, to do it. I never use a result from this paper, as I derive the LO stability differently, making this formulation of the problem irrelevant to the thesis. I do want to note somewhere though that this is a viable way of obtaining a LO phase stability requirement

Due to the balanced nature of the beam splitter, the classical intensity noise of the LO is not present in the BHD signal. This and the ability to set the homodyne angle make BHD ideal for measuring quantum states of light. For example, BHD can be used to detect the level of squeezing in a quadrature of the signal beam by setting the homodyne angle to the quadratures angle (e.g. [11]).

BHD is useful for inferring the quantum state of two-photon phenomena, such as squeezing¹, as without it the phase quadrature cannot be measured. A two-photon process can be described in terms of the amplitude and phase quadrature operators, denoted as $\hat{X}$ and $\hat{Y}$ respectively (e.g. [12]). These operators are related to the

¹and photon bunching

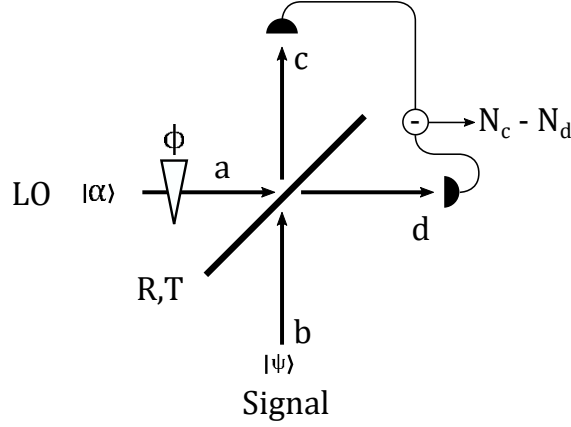


Figure 5.1: A coherent state $|\alpha\rangle$, enters port a of a beam splitter. A signal $|\Psi\rangle$ enters port b . These two beams interfere, resulting in photons being detected at the c and d ports. The two photo-currents are subtracted, and the signal that is produced is proportional to an arbitrary quadrature of the state $|\Psi\rangle$. The quadrature that is measured is determined by the homodyne angle, ϕ . This is expressed mathematically by equation (??).

annihilation and creation operators of the electromagnetic field, $\hat{a}^\dagger(t)$ and $\hat{a}(t)$, by

$$\hat{\mathbf{X}}(t) = \frac{1}{2} (\hat{a}(t) + \hat{a}^\dagger(t)) \quad (5.1)$$

$$\hat{\mathbf{Y}}(t) = \frac{1}{2i} (\hat{a}(t) - \hat{a}^\dagger(t)) \quad (5.2)$$

$$\hat{\mathbf{X}}_\phi = \hat{\mathbf{X}} \cos \phi + \hat{\mathbf{Y}} \sin \phi. \quad (5.3)$$

Consider a LO field $\hat{a}$ and a signal field $\hat{b}$. It is arranged so that these fields interfere on a beam splitter are detected on two photodiodes, c and d . The fields

these diodes detect are $\hat{c}$ and $\hat{d}$. The photo-currents are then

$$\hat{\mathbf{N}}_c = \hat{c}^\dagger \hat{c} = \left(\sqrt{R} \hat{a}^\dagger + \sqrt{T} \hat{b}^\dagger \right) \left(\sqrt{R} \hat{a} + \sqrt{T} \hat{b} \right) \quad (5.4)$$

$$= R \hat{a}^\dagger \hat{a} + T \hat{b}^\dagger \hat{b} + \sqrt{RT} (\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}) \quad (5.5)$$

$$\hat{\mathbf{N}}_d = \hat{d}^\dagger \hat{d} = T \hat{a}^\dagger \hat{a} + R \hat{b}^\dagger \hat{b} - \sqrt{RT} (\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}). \quad (5.6)$$

For BHD, the beam splitter must be balanced so $R = T = 1/2$. This results in the difference in the photo-currents being

$$\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d = \hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}. \quad (5.7)$$

The operators $\hat{a}$ and $\hat{b}$ can be ‘linearised’, $\hat{a} \rightarrow \alpha + \delta a$, [?] in order to separate the quantum behaviour and classical behaviour. Furthermore, the phase between the signal LO field can be expressed in terms of the homodyne angle with $\hat{b} \rightarrow \hat{b} \exp(i\phi)$. By neglecting terms of the order $\delta a \delta b$, this allows one to rewrite equation (5.7) as

$$\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d = 2\alpha\beta \cos \phi + 2\alpha\delta \mathbf{X}_{-\phi}^b + 2\beta\delta \mathbf{X}_\phi^a. \quad (5.8)$$

The variance, $\sqrt{V(X)} = \Delta X$, of this measurement would then be

$$V(\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d) = 4\alpha^2 V(\delta \mathbf{X}_{-\phi}^b)^2 + 4\beta^2 V(\delta \mathbf{X}_\phi^a)^2. \quad (5.9)$$

If the LO is much stronger than the signal field, $\alpha \gg \beta$, then the measurement is

$$\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d = 2\alpha\beta \cos \phi + 2\alpha\delta \mathbf{X}_{-\phi}^b, \quad (5.10)$$

which is proportional to general quadrature of the signal.

5.2 BHD versus DC readout for A+

There are some advantages for BHD over DC readout [13], although for A+ a quantifiable difference is hard to make. The main attraction for implementing BHD is the ability to reduce the dark fringe offset that is required to produce the LO for DC readout. This offset gives rise to most of the limitations associated with DC readout, such as the coupling between the SRC length noise and the DARM signal. BHD also provides the ability to select which quadrature of the signal beam is measured which is a requirement for novel interferometer configurations.

The current detection scheme is known as DC readout [14]. A differential offset in the arm lengths produces the LO in DC readout. The LO beats with a signal created by a gravitational wave in order to produce intensity modulations at the AS port. The size of the LO can be controlled with the DC offset, and the homodyne angle can be controlled with a differential arm power. This method of readout has some key positive outcomes, such as there being no phase noise between the LO and the signal, and the LO and signal being perfectly aligned and mode matched. The problems with this readout technique are associated with requiring the DC offset as this light gives rise to noise couplings.

For BHD to be worthwhile it must be at least as good as DC readout. Thus, BHD should be considered in relation to the perks of DC readout. Unlike DC readout,

BHD may introduce phase noise between the LO and signal[10]. Therefore the relative stability between the LO and signal beams has to be considered. Control over the alignment and mode matching between the LO and the signal is also important. LO light that is scattered into higher order modes, by misalignment or mode mismatch, may backscatter from the OMC into the interferometer and could result in noise in the DARM signal. The implementation of BHD will only be successful if it provides a stable enough LO beam; sufficient mode-matching between the LO and signal beam, and active alignment of the LO and signal beams.

BHD has some practical benefits over DC readout[13]² as well as some interesting prospects in relation to the quantum noise limit of the detector. The practical benefits arise due to being able to eliminate the DC light required for DC readout to work. The primary benefit is in the reduction in SRC length noise coupling to the gravitational wave signal. Another benefit is being able to reduce the scattering from the septum window between HAM5 and HAM6. Quantum noise shaping is, along with other interferometer enhancements (e.g [5]), related to the complete free choice of homodyne angle that a BHD readout scheme permits.

²One ‘advantage’ that is often talked about in relation to BHD is based on the electronic noise of the readout photodiodes. This is not a real advantage. aLIGO currently uses DC readout and is not limited by the electronic noise of the transimpedance amplifier. As the BHD signal is obtained through subtracting the signals from two photodiodes, it is possible to obtain a high SNR when designing the transimpedance amplifier by subtracting the currents produced by the photodiodes rather than having two separate transimpedance amplifiers and subtracting the signals made by these later. This is not really an advantage, as low enough electronic noise can be obtained using other construction techniques that allow for small DC signals while still having signal shaping that gives a large enough SNR in the detection band. Having separate transimpedance amplifiers for each photodiode also has useful diagnostic advantages, such as being able to see the sum of the two photodiode signals. Therefore, direct subtraction of the photodiode signals should not be considered an advantage of BHD over DC readout.

One of the practical outcomes of implementing BHD is the potential to reduce SRC length noise coupling. Noise on the length of the SRC can couple to the gravitational wave signal via two routes. The dominant coupling, which is proportional to the DC offset, will arise due to the power entering the SRC. The other coupling arises due to the optical spring between the SRC and the arm cavity length.

The dominant coupling due to the power entering the SRC can be intuitively understood by considering a Fabry-Perot cavity. For small changes in cavity length, the light transmitted by a cavity that is on resonance or anti-resonance will be modulated by a term proportional to the square of the length change. This signal due to the length of the SRC is noise. The amplitude of this noise is linearly dependent on the amount of light entering the cavity which is a result of the DC offset. By decreasing the DC offset, this noise is decreased. As this noise is in the amplitude quadrature of the signal beam, unlike the gravitational wave signal which is in the phase quadrature, it will not be amplified by the LO if the homodyne angle is set to readout from the phase quadrature.

The second route SRC length noise may couple to the signal is via the optical springs between the SRC and arm cavities. Due to this optical spring effect, a change in length of the SRC will cause the arm cavity lengths to change as well [15]. The effect of optical springs on the SRC length noise coupling strength is much smaller than the first effect, so the estimate for the reduction of SRC length coupling will only considering the first effect.

SRC length noise is currently too high to meet the A+ requirements [16]. To estimate the reduction in SRC length noise that BHD may provide, the amount of

DC offset that is currently used should be compared to the amount that is proposed for BHD. aLIGO has 23 mW of light due to the DC offset and 1 mW of residual light (light that cannot be removed) at the AS port [17]. The AS power for A+ is between 1 and 5 mW. Therefore, a conservative estimate for the factor that the SRC length noise coupling is reduced by is four.

Strictly speaking, these are not properties of BHD readout as scatter from the septum window and the SRC length noise could be reduced by other means. For instance, the septum window could be removed and suspensions with additional stages could be used. However, these solutions are not trivial to realise, and they are no more practical than implementing BHD. The septum window is a key part in being able to carry out commissioning tasks within HAM6. The number of times HAM5 and the other connected chambers are vented should be kept low as each time this is done gas and water vapour enters the system (this contributes to the noise floor of the interferometer). This residual gas and vapour will not be removed as this is a large engineering challenge. The design and install of three new suspensions is also a difficult task, which ultimately gives less benefit compared to a BHD readout scheme. Realistic upgrades that will reduce to the SRC length noise coupling are currently being developed [18].

Different gravitational wave sources carry their information in different frequency bands and the detector may be optimised to reflect this [19, 20, 21, 5]. For instance, BNS signals continue to higher frequencies than BBH signals so it may be of interest to optimise the detector around these frequencies to improve our understanding of the properties of BNS systems. To optimise the sensitivity of the interferometer to

different sources the detectors noise would need shaping, a task that can be achieved, along with other upgrades, BHD.

The current generation of gravitational wave detectors are going to be limited by a combination of radiation pressure noise and shot noise in their detection bands. These two noise sources are intrinsically linked by the Heisenberg uncertainty principle and can both be understood as arising due to the number of photons used to probe the position of the end test masses. This limit in the noise floor is known as the standard quantum limit [5]. To circumvent this limit the homodyne angle must be free to choose; a functionality that is provided by BHD. In principle, DC readout gives the ability to set the homodyne angle [13], although in practice it is much less flexible than a BHD configuration.

A squeezed light field can be used to surpass the standard quantum limit, and to detect these squeezed states the homodyne angle must be correctly set. The exact way in which the quantum noise is changed is controlled by the squeezing angle. Currently the squeezing angle is set to be at the same angle as the readout quadrature, rather than the preferred option of setting the readout quadrature to the squeezing angle. The effect that controlling the squeezing angle has on the quantum noise floor has been demonstrated at LLO [1]. However, [1] used a diagnostic BHD set-up which was not intended for providing the low noise required to make gravitational wave detections.

More advanced upgrades, e.g. see [5], could alter quantum noise substantially, and they rely on the implementation of BHD. These changes to the interferometer are nontrivial, and would require significant redesign of many parts of the layout, so the

implementation of BHD for A+ does not necessarily correspond to a suitable BHD system for one of these more advanced interferometer configurations. However, BHD for A+ would provide experience with implementing BHD on a large scale interferometer, which would help when considering how to implement these more novel interferometer configurations.

In summary, BHD has a couple of key advantages over DC readout. BHD allows for the reduction in SRC length noise coupling and, in conjunction with other upgrades, the ability to shape the quantum noise floor of the interferometer. In DC readout, the SRC length coupling comes from the DC offset required to provide a LO beam for the signal generated by a differential modulation in arm length to beat against. This DC offset is not required for BHD. Most interferometer configurations that beat the SQL assume that the homodyne angle can be freely set. The drawbacks of BHD when compared DC readout are all engineering challenges; nothing fundamentally favours DC readout over BHD.

5.3 Local Oscillator Path Length Stability Requirement for A+

This Section will go through the steps needed to derive the phase noise requirement for the beam path of the A+ LO. This uses a semi-classical approach, and the end result is the same as if a full quantum mechanical treatment was used. The quantum mechanics of this problem is included in the A+ sensitivity curve and the expected power in the signal beam. The phase noise of the LO must be kept low enough so that it does not result in intensity fluctuations greater than those due to the shot

noise of the light in the signal beam. This requirement was also derived in [22], [23] and [24], however in this work the most current layout was considered.

Phase noise arises due to phase fluctuations in the LO beam being indistinguishable from phase fluctuations in the signal beam. A gravitational wave produces a phase modulated signal in the signal beam, and the LO field phase is modulated by the motion of the mirrors in the LO path. When these beams are combined, it is the phase difference between the two fields that produces a signal, so the phase noise of the LO path must be low enough that the signal that is generated by the interference of the LO and signal beams is predominantly due to the phase modulation of the signal beam.

The relationship between the differential phase in the arms and the power in the signal beam was used to find the power noise for the beam exiting the SRM in A+. The linear response function due to a differential phase change in the arms was computed with FINESSE. This was used to deconvolve the strain sensitivity curve to produce a power noise curve for the light exiting the AS port of the interferometer. Since the interferometer is linear, noise is transformed linearly. This means that the quantum nature of the noise is irrelevant to the calculation.

This light noise curve was then used to set the maximum allowable intensity fluctuations due to phase noise between the LO and the signal beams. This phase noise can then be converted into a distance, and this produces a stability requirement for the steering optics used to combine the LO and signal beams.

The following Sections will go through the aforementioned steps. First the LO path will be described. Then a linear response function between a differential strain and signal at the AS port will be obtained via FINESSE. Next, The sensitivity curve for A+ [25] will be expressed in units of $\frac{W}{\sqrt{Hz}}$. This will be converted into a longitudinal motion requirement. Finally, there is a short discussion about the proposed optics that fulfil this requirement.

5.3.1 Local Oscillator Path

The beam path for the LO and signal beams up to the point of recombination will be described in this Section. The LO will be picked off from the power recycling cavity and guided by one steering mirror, BHDM1, to the BHD combining beam splitter, BHDBS2. The signal beam will exit the OFI and be steered by one mirror, OM0, onto BHDBS2. OM0 and BHDM1 will be used to align the beams so they overlap on BHDBS2. The angle of BHDBS2 will be used to make the LO and signal beams overlap in the far field.

The LO could have been derived from a separate laser, but to acquire laser light with low frequency and intensity noise the LO will be picked off from a point within the interferometer. Two options for the LO pick off were considered [22]: a beam picked off from the main beam splitters AR coating (BSAR), or the beam transmitted through PR2 (POP). The determining factor in choice of pick off point was the effect of backscatter. The POP port was chosen as it is far superior to the BSAR choice at all frequencies and scatter phases. This is because backscattered light from the BSAR enters only one arm, thus making it a differential affect. Any backscatter that enters the POP port is common-mode, and therefore DARM is far less sensitive

to this source of backscatter.

The layout for the optics which combine the LO and signal beam is shown in Figure 5.2. The beam transmitted by PR2 is used for control of the interferometer, so BHDBS1 is needed to pick off some of this light. The LO beam is steered towards the combining beam splitter BHDBS2 with BHDM1. Two lenses (**need names for these**) are used for mode matching the LO beam to the signal beam. The signal beam is transmitted through the SRM and OFI, and then it is steered towards BHDBS2 with OM0.

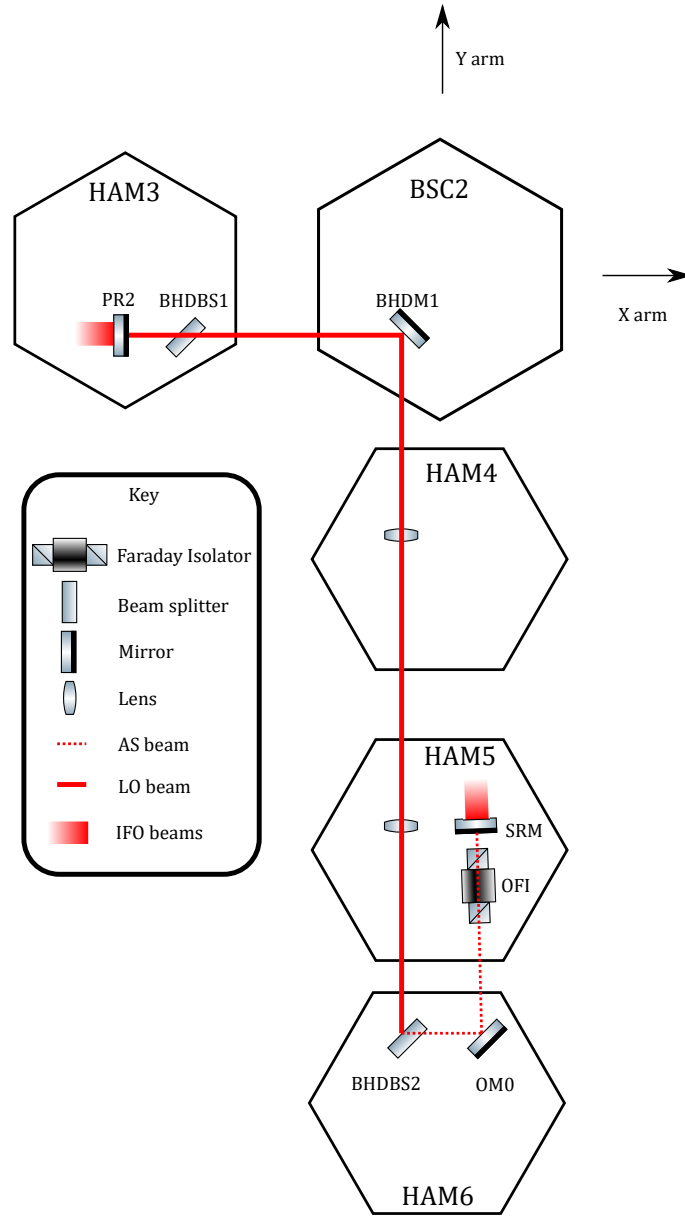


Figure 5.2: Sketch of the LO and signal beam paths. The HAM tables are shown by the hexagons. Most of the interferometer has not been shown for clarity. The main interferometer beam is indicated by a thick, gradated red line. The thinner red line indicates the LO beam, and the signal beam is shown with a dashed red line.

5.3.2 The Linear Response Function Between Differential Strain and Power at the AS Port

The linear response function of the A+ detectors was obtained using FINESSE. Since the core optics of the interferometer will not be changing in a way relevant to this calculation, the LLO model [26] was used to calculate the response between differential motion of the ETMs and the power transmitted through the OMC. The linear response function at the frequencies contained in the detection band are not influenced by the OMC. This means no major modifications to the FINESSE model were needed, and so the model represents a state that the interferometer would be in for observing gravitational waves.

FINESSE allows one to measure linear response functions by applying a ‘swept sine’ signal to a component and demodulating the light at the frequency of the applied signal. To create a differential arm length signal with FINESSE, `fsig` commands were used to apply two signals in anti-phase with each other to the ETMs,

```
fsig sig1 ETMXHR 1 0
fsig sig2 ETMYHR 1 180.
```

The frequency of this modulation was swept, and the signal produced at the AS port was measured by demodulating the light at the modulation frequency. The commands

```
xaxis sig1 f log 5 5000 2999
put P_DC_OMC_probe f1 $x1
```

do this.

For this calculation, the linear response function measured in FINESSE needs to be converted into different units. To get the linear response function between arm motion and AS power into units of $\frac{\text{Power}}{\text{Strain}}$, or $\frac{\text{W}}{\text{h}}$, several factors are required. Since the input modulation is doubled by moving the mirrors differentially, the response needs dividing by two. To account for the ETMs retro-reflecting the light, the response needs dividing by two again. To convert this from phase units to strain units, a factor of $\frac{\lambda}{2\pi L_{\text{arm}}}$ is needed. Therefore, the output of the FINESSE simulation needs to be multiplied by $\frac{1}{2} \frac{1}{2} \frac{2\pi L_{\text{arm}}}{\lambda}$ to be in units of $\frac{\text{W}}{\text{h}}$. The linear response function between a differential arm strain and the power at the AS port is shown in Figure 5.3.

5.3.3 Converting the Sensitivity Curve for A+ into a Longitudinal Motion Requirement

To convert the A+ sensitivity curve into a path length stability requirement for the LO, a few steps are required. First, the sensitivity curve must be expressed in units of $\frac{\text{W}}{\sqrt{\text{Hz}}}$. Second, the intensity noise of the signal beam needs to be converted into a phase noise. In order for the phase noise of the LO to not limit the performance of the gravitational wave detector, the phase signal in the signal beam must be greater than the phase noise of the LO beam. Finally, a factor of ten safety margin is required. The HRTS requirement is then obtained by including the number of reflective optics in the LO path and signal path.

The A+ sensitivity curve was multiplied by the function shown in 5.3 to convert it from units of $\frac{\text{h}}{\sqrt{\text{Hz}}}$ to $\frac{\text{W}}{\sqrt{\text{Hz}}}$. This is shown in Figure 5.4. At frequencies above 1 kHz the requirement is flat as this is the region dominated by shot noise. At lower

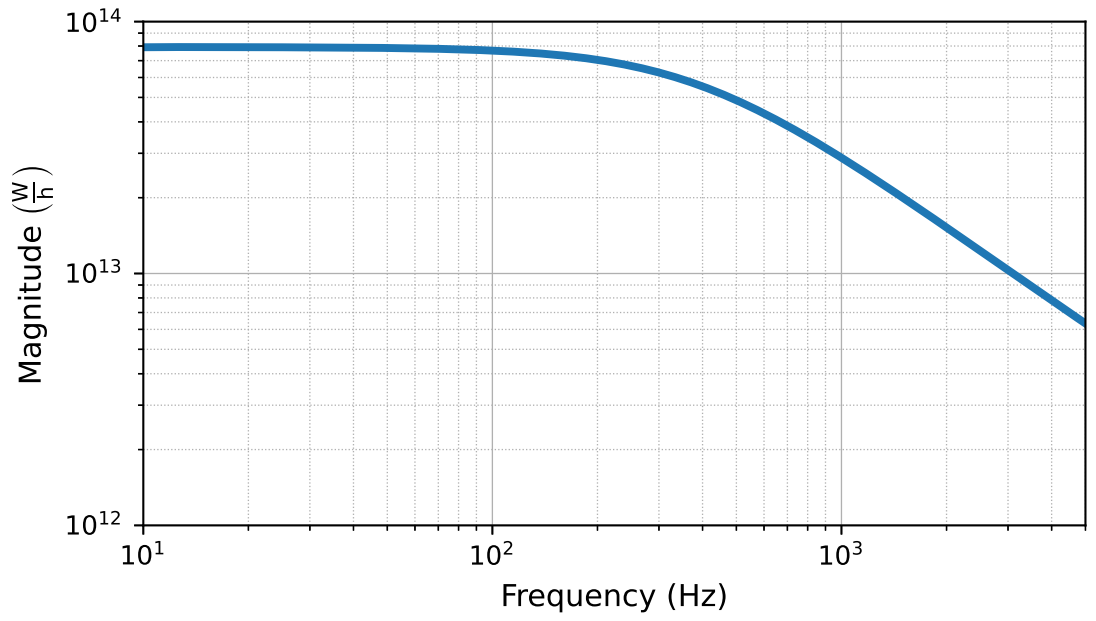


Figure 5.3: The linear response function between strain and light power, expressed in units of Light Power/gravitational wave strain, at the AS port was obtained from a FINESSE simulation of the LLO detector. The corner frequency at ≈ 300 Hz comes from the pole of the coupled SRC-arm cavity.

frequencies the requirement is relaxed as the sensitivity curve becomes dominated by other noise sources.

Next, a change in power measured in the combined LO/signal beam due to a phase modulation, i.e. path length modulation, in the LO or signal path needs calculating. When the LO and signal beams interfere at BHSBS2, the exiting beams have power P_c and P_d . These are given by

$$P_c = \left| \frac{1}{\sqrt{2}} E_s e^{i\omega t} + \frac{1}{\sqrt{2}} E_{\text{LO}} e^{i\omega t + \phi} \right|^2 = \frac{|E_s|^2}{2} + \frac{|E_{\text{LO}}|^2}{2} + |E_s E_{\text{LO}}^\dagger| \cos(\phi) \quad (5.11)$$

and

$$P_d = \left| \frac{1}{\sqrt{2}} E_s e^{i\omega t} - \frac{1}{\sqrt{2}} E_{\text{LO}} e^{i\omega t + \phi} \right|^2 = \frac{|E_s|^2}{2} + \frac{|E_{\text{LO}}|^2}{2} - |E_s E_{\text{LO}}^\dagger| \cos(\phi), \quad (5.12)$$

where E_s and E_{LO} are the (complex) field amplitude of the signal and LO beams; ω is the laser frequency; t is time and ϕ is the homodyne angle. The difference in these signals gives BHD signal

$$P = 2|E_s E_{\text{LO}}^\dagger| \cos(\phi). \quad (5.13)$$

Differentiating with respect to ϕ gives the relationship between a change in power and a change in phase between the two beams,

$$dP = -2|E_s E_{\text{LO}}^\dagger| \sin(\phi) d\phi, \quad (5.14)$$

and since the homodyne angle is 90° this becomes

$$dP = -2|E_s E_{\text{LO}}^\dagger| d\phi. \quad (5.15)$$

To convert ϕ to distance, use

$$\frac{\lambda}{2\pi} d\phi = dx. \quad (5.16)$$

The light noise curve shown in Figure 5.4 should be multiplied by $\frac{\lambda}{2\pi} \frac{1}{2|E_s E_{\text{sig}}^\dagger|}$ to obtain the noise floor of the interferometer in units of LO path length noise.

To determine the longitudinal motion requirement for the optics that combine the LO and signal beam the number of optics and, since there are many uncorrelated noise sources in LIGO, a factor of ten safety margin needs to be included. The requirement assumes, based on the current design, that there will be two mirrors at 45° which are used to steer the AS and LO beams. As the optics reflect at 45° , the path length change due to the optics motion is multiplied by $\sqrt{2}$. The beams are combined on a beam splitter, and the motion of this beam splitter needs to be counted twice. Figure 5.5 illustrates this.

For a 98 mW LO beam and a 5 mW signal beam, the noise floor of the interferometer in units of LO path length is shown in Figure 5.6. The requirement for the longitudinal motion of optics in the LO or signal path at 100 Hz is $5 \times 10^{-17} \text{m}/\sqrt{\text{Hz}}$.

To determine the amount of isolation required, the typical motion of the optical table needs to be compared to the longitudinal motion requirement for the optics. A typical motion of the optical benches at 10 Hz is $10 \times 10^{-1} \text{m}/\sqrt{\text{Hz}}$ [27]. To achieve this with passive isolation, triple suspensions will be used. These suspensions are

known as HAM Relay Triple Suspensions (HRTS)[28].

5.3.4 Summary of LO path length Stability

Details of the LO path were discussed and analysed. The LO will be derived from a beam picked off from the POP port, and a sketch of how the signal and LO will be combined is shown in Figure 5.2. Starting with the linear response function between a differential strain and a signal at the AS port and the strain sensitivity curve for A+, we were able to derive a relative path length stability requirement for the LO and signal beams. At 100 Hz this requirement is $5 \times 10^{-17} \text{m}/\sqrt{\text{Hz}}$. By comparing this motion to a typical motion of the optical table, we were able to derive an isolation requirement, shown in Figure 5.6. This longitudinal phase noise requirement led to triple suspensions being chosen to steer the LO and signal beams onto the combining beam splitter.

5.4 Conclusion

One of the major changes to LIGO for A+ will be the implementation of BHD. Combined with frequency dependent squeezing, BHD offers the possibility to optimise the detector for different gravitational wave sources. Along with this, BHD offers a way by which the coupling between the SRC length noise and DARM can be reduced.

BHD presents some technical challenges over DC readout. The LO phase noise must be kept low enough as to not contaminate the signal at the AS port with LO path noise. The requirement for the phase noise is shown in Figure 5.6. This

requirement can be met by using triple suspensions to deliver to LO and signal beams to the beam splitter they are combined on.

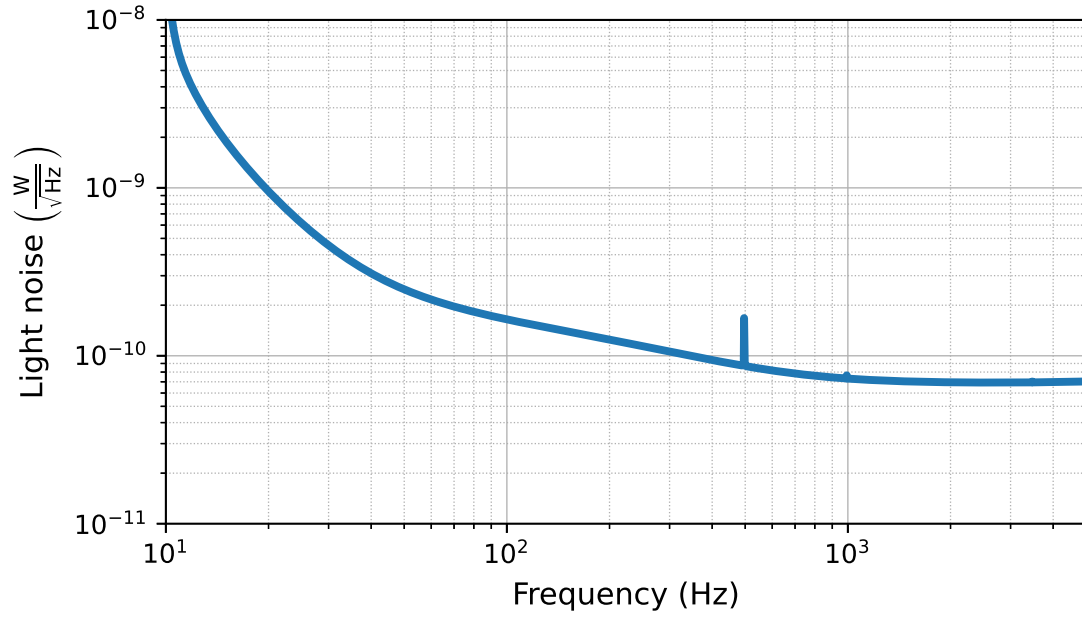


Figure 5.4: The intensity noise of the light at the AS port for the A+ detector. This was obtained by deconvolving the detectors sensitivity curve with the function shown in Figure 5.3. At frequencies above 1 kHz the noise is flat as it is dominated by shot noise. At lower frequencies the noise is increased relative to shot noise due to radiation pressure, coating thermal noise and other noise sources. The spike in the spectrum at 500Hz represents the presence of suspension modes (actually find out what this is).

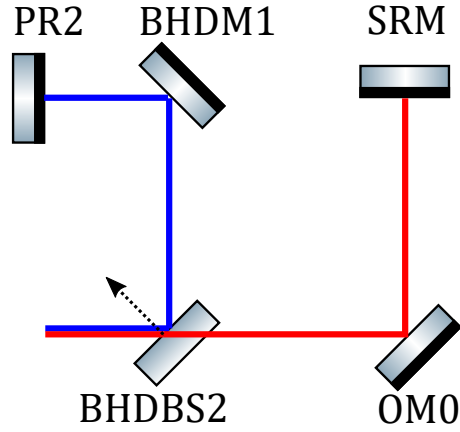


Figure 5.5: A sketch of the LO and signal beam paths. The LO beam path is shown in blue, and the signal beam is shown in red. A dashed arrow indicates the direction that the recombining beam splitter, BHDBS2, may move in. If BHDBS2 moves in this way, the phase of the signal beam has increased while the phase of the LO beam has decreased at the point where the signal and LO beams combine. This means that the recombination beam splitter motion needs to be counted twice in calculations of the phase noise requirement.

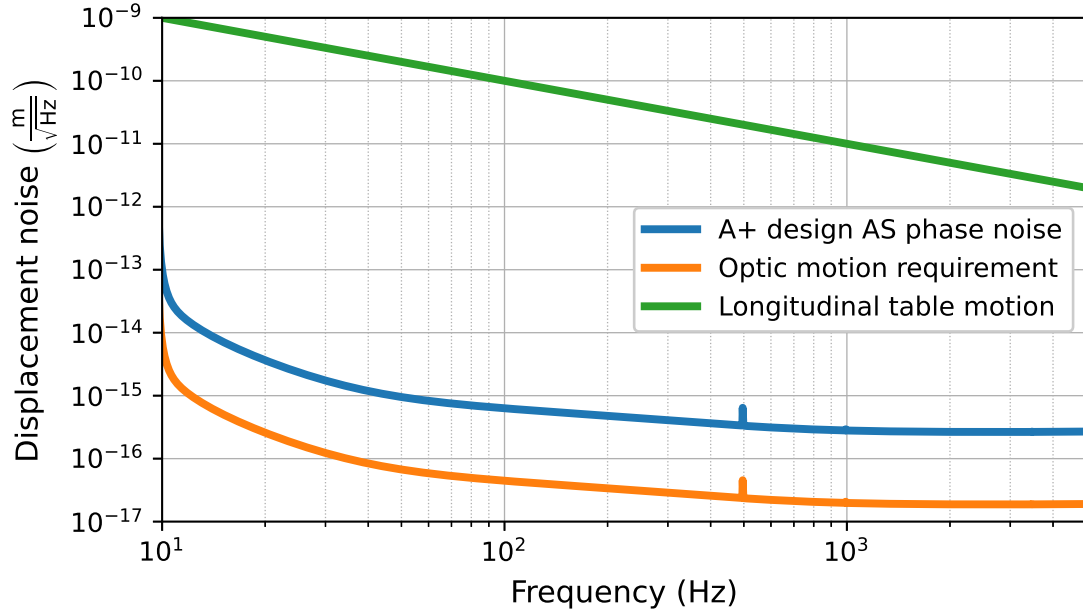


Figure 5.6: The phase noise between the LO and the signal beams causes noise. The sensitivity curve for A+ sets the limit on how much path length noise is acceptable between the AS and LO beams. The blue line shows sensitivity curve is shown in units of path length noise between the AS and LO beams, and the orange line shows the requirement of HRTS mirror motion. The table motion was modelled as $1/f$ noise.

Chapter 6

Design of the LIGO A+ HAM6 layout

Fundamentally, the optical layout in HAM6 must combine the signal beam and the LO beam and sense the combined beams to form a balanced homodyne detector. The implementation of BHD for LIGO A+ requires a complete redesign of the HAM6 layout. This chapter is comprised of three sections. Section 6.1 is a description of the functionality and design constraints that all candidate HAM6 layouts must satisfy. In Section 6.2, the optical properties of the proposed HAM6 layout are examined. Section 6.3 is an analysis of how mode mismatches between the interferometer and the OMC can be corrected by the Active Wavefront Control (AWC). Section 6.4 is the conclusion for this chapter. This work was undertaken by a team including myself, Ken Strain, Russell Jones, Stephen Webster, Mark Barton and Eric Oelker.

6.1 The Constraints and Requirements for the Optical Layout Within HAM6

This section will discuss the functionality required from the optical layout within HAM6 and describe the practical constraints imposed by the existing hardware and infrastructure. HAM6 must contain an optical layout that combines the LO and signal beams as well as providing steering and mode matching between the interferometer and the OMC. There are a few properties of the HAM6 layout that cannot be changed such as the entry point of beams and size of the optical table. The main design challenge was achieving 45° of Gouy phase between the optics for mode matching as this requires a lot of space relative to the size of the optical table.

6.1.1 Constraints

As the constraints give a starting point for all designs, they will be discussed first¹. These include: where beams enter the HAM6 chamber, the size of the nominal mode entering HAM6 and the size of the OMC waist. The size of the HAM optical table does restrict the design, however table extensions can be made depending on where they would be needed.

¹It could be argued that requirements are a constraint in a design. Here, the constraints are geometric properties which we have no freedom in, such as the input beam shape, the OMC geometry and the layout of the rest of the interferometer. A requirement, such as the desired minimum mode matching, is a property which does not have a set ‘physical form’. For instance, while Gouy phase does correspond to a physical distance, not all 45° Gouy phase spaces are the same physical size.

6.1.1.1 Beam Entry Points

The entry point of the signal and LO beams in principle can be anywhere in HAM6; however, in reality they are fixed due to the location of HAM6 with respect to the interferometer; the location of the septum windows between HAM5 and HAM6; and due to the many other components of the interferometer that the signal and LO beams need to avoid. The LO beam is steered with BHDM1 onto BHDBS2, and the signal beam is steered with OM0 onto BHDBS2. These components are shown in Figure 5.2. This is required to provide the degrees of freedom to co-align the signal and LO beams. There is a freedom in choice for the locations of OM0 and BHDBS2, but there is no choice in the entry points into HAM6 for the signal and LO beams.

6.1.1.2 Input and Target Beam Parameters

The nominal beam parameter at the SRM is $q_{\text{SRM}} = -3.55 + 1.09i \text{ m}$ [29], so the design of the HAM6 mode matching optics is based on this number. As discussed in Chapter 7, the SRM beam parameter could differ from its nominal value. Chapter 7 describes how geometry errors in the SRC cause the arm cavity and the SRC to be mismatched. This means that the radius of curvature and beam size at the SRM will not correspond to the eigenmode of the SRM, and the beam parameter at the SRM that should be considered corresponds to the arm cavity mode. This set of likely modes, shown in Figure ??, was taken into consideration when choosing a layout for HAM6 since candidate layouts may have been more or less robust to beam parameter errors at the SRM.

The current OMC design, with trivial modifications, will be used to remove junk light from the signal beam, so there is no choice in target waist size. The nominal

OMC waist size is $485\text{ }\mu\text{m}$. The beam enters the OMC through a flat mirror, and the waist is located 140 mm from this mirror. There is a negligible amount of astigmatism inherent in the design of the OMC. Mode matching optics therefore have to match the nominal SRM and OMC modes.

6.1.1.3 Equal Path Lengths after BHDBS2

It was decided that the two beams after BHDBS2 would have the same path length to reduce the cost of the HAM6 optics. This may also improve the common-mode noise rejection ratio as asymmetries may enhance the effects of noise couplings. To achieve high levels of common mode noise rejection, both paths should be as similar as possible. Only one path needs to have imperfections for the common mode rejection ratio to be degraded [7]. For instance, it may be a bad design choice to have a path with a mirror whose only purpose is to fold the layout so that a slightly better readout platform can be used. However, it should be noted that the amount of common mode rejection required is related to the classical noise of the laser and its shot noise limit. Since the LO in LIGO A+ will be derived from the PRC it will already be stable (shot noise?), so little CMRR will be needed. In addition to this, the layout will be in a clean environment with a low amount of residual gas; thus reducing mechanisms whereby imbalances due to path length difference can generate noise (may remove?). Therefore, the trade-off between mirror number and path length symmetry versus readout platform preference is not necessarily easy to evaluate.

6.1.2 Requirements

The HAM6 layout is required to provide steering and mode matching for the signal-LO beam into the OMC. The allowable mode-mismatch is 2%. The losses due to components in HAM6 is summarised in Table 6.1[30]. After BHDBS2, the layout must have a pair of mirrors separated by 90° and another pair separated by 45° of Gouy phase to provide orthogonal control over the beam pointing and mode-matching.

6.1.2.1 Mode Matching Losses

The goal for mode matching between the arm mode and the OMC mode is $> 98\%$. The SAMS will be used to correct for mode mismatch; they will provide ± 100 mD of optical power tuning range with a bandwidth of the order 100 Hz. Astigmatism degrades the mode matching, and non-zero angles of incidence (AOI) of beams onto curved optics cause the reflected beam to become astigmatic [31]. This prevents designs using large AOI.

Nominally, the role of the AWC in HAM6 is to correct for geometry errors of the optics after the SRM such that the mode matching between the arm mode and the OMC mode is maximised. This assumes that changes in the interferometers mode, for instance due to a change in thermal state, will be corrected for by other subsystems e.g. the TCS. In practice, operators and commissioners use each tunable aspect of the interferometer to improve its sensitivity to gravitational waves. For instance, it has been found that TCS can be used to decrease laser intensity noise coupling to DARM. The optical layout and AWC functionality in HAM6 were designed with this flexibility in mind.

6.1.2.2 Gouy Phase Considerations

While in principle only two curved mirrors in each path to the OMC would be required to mode match the beam from the interferometer to the OMC, even a single mirror may be able to do that, a lens in the OFI will be used to provide further Gouy phase separation between the two active mode matching mirrors in each path after BHDBS2. An active lens was considered here, but concerns of OFI assembly heating [32] and the unavailability of flint glass meant that this idea was discarded. As well as this, the range of optical powers for the SAMS is large enough that it would be an over-design to implement an active lens in the OFI.

To provide orthogonal mode matching control, two of the mirrors should be separated by 45° . This Gouy phase separation requirement gives us a method of generating candidate designs, as this can be interpreted as needing the mode matching mirrors to be as far apart as possible from each other on the HAM table. As the Rayleigh range of the beams is on the order of one meter, and most Gouy phase is accumulated within one Rayleigh range either side of a beam waist, a beam will have to travel a significant distance across the table before it acquires sufficient Gouy phase.

Additionally, to provide orthogonal alignment control for each beam path after BHDBS2, two of the mirrors in each of the said beam paths are required to be separated by 90° of Gouy phase. For all designs, the two mirrors for alignment were always immediately after BHDBS2 and immediately before the OMC. It was found that placing the first mirror close to BHDBS2 often provided good layouts.

6.1.2.3 Readout Platform Suspension and Auxillary Components

The OMCs are expected to be the largest contributor of back-scatter from HAM6 to the interferometer. By having the OMCs after the signal and LO beams are combined and by having the OFI protect the interferometer, the requirement on the motion of the OMCs is fairly lax. Nevertheless, a single stage suspended platform is required. This is known as the readout platform suspension (RPS). This will provide isolation down to 4 Hz. Both OMCs will need to be on this platform, meaning all designs will need to have the OMCs fairly close to each other. Too large an RPS is mechanically infeasible as it will reduce space for other auxiliary components.

Additional sensors in HAM6 are required for controlling various degrees of freedom within the layout. Some of these degrees of freedom will be monitored using beams which are picked off from the main beam path, so additional mirrors will be required to steer these beams around the chamber. A set of smaller optics will be installed to provide diagnostic beams which can be imaged with CCDs. A shutter will be required to protect the hardware in HAM6 from being damaged by the large amounts of light that can potentially be transmitted through the SRM. Appendix D describes how the longitudinal and pointing degrees of freedom associated with BHD may be sensed and controlled.

6.1.3 Summary

In summary, the main constraints for the design of HAM6 are where the beams enter the chamber; the size of the optical bench; the input mode and the OMC design. There must be low amounts of loss; therefore, layouts with large AOI onto curved

mirrors should be avoided. The optical layout within HAM6 constitutes a nontrivial design task as it is desirable to have orthogonal control over the mode matching and beam direction. This requires the optical axis to be of a similar length to the maximum spacing permitted by a HAM table. In addition to this, the HAM6 layout needs to leave space for all the other sensors and beams used for control. This means that all designs will use nearly all of the table space.

HAM6 loss Source	Target for O5
OMC throughput	$< 2\%$
OM3 transmission (or equivalent) for sensing	1–2%
PD quantum efficiency	1%
Mode mismatch to the OMC	$< 2\%$ (as low as possible)
Any other optical loss	$< 1\%$

Table 6.1: Target for losses in HAM6. Reproduction of [30]

6.2 The Chosen HAM6 Optical Layout

The design that is being taken forward at the time of writing is shown in Figure 6.1. Optically, there are many layouts which could have satisfied the requirements and constraints laid out in Section 6.1. The choice of layout for HAM6 was ultimately determined by mechanical factors such as table space, number of suspensions and RPS design. The optical properties of this layout are summarised in Table 6.2. The mirrors OMA1, OMB1, OMA2, OMB2 etc. will be referred to as OMx1 and OMx2 as, usually, it is not important to make a distinction between the A and B paths.

The beam size and Gouy phase as a function of position along the optical axis is shown in Figure 6.2. Significant points along the optical axis are also shown in

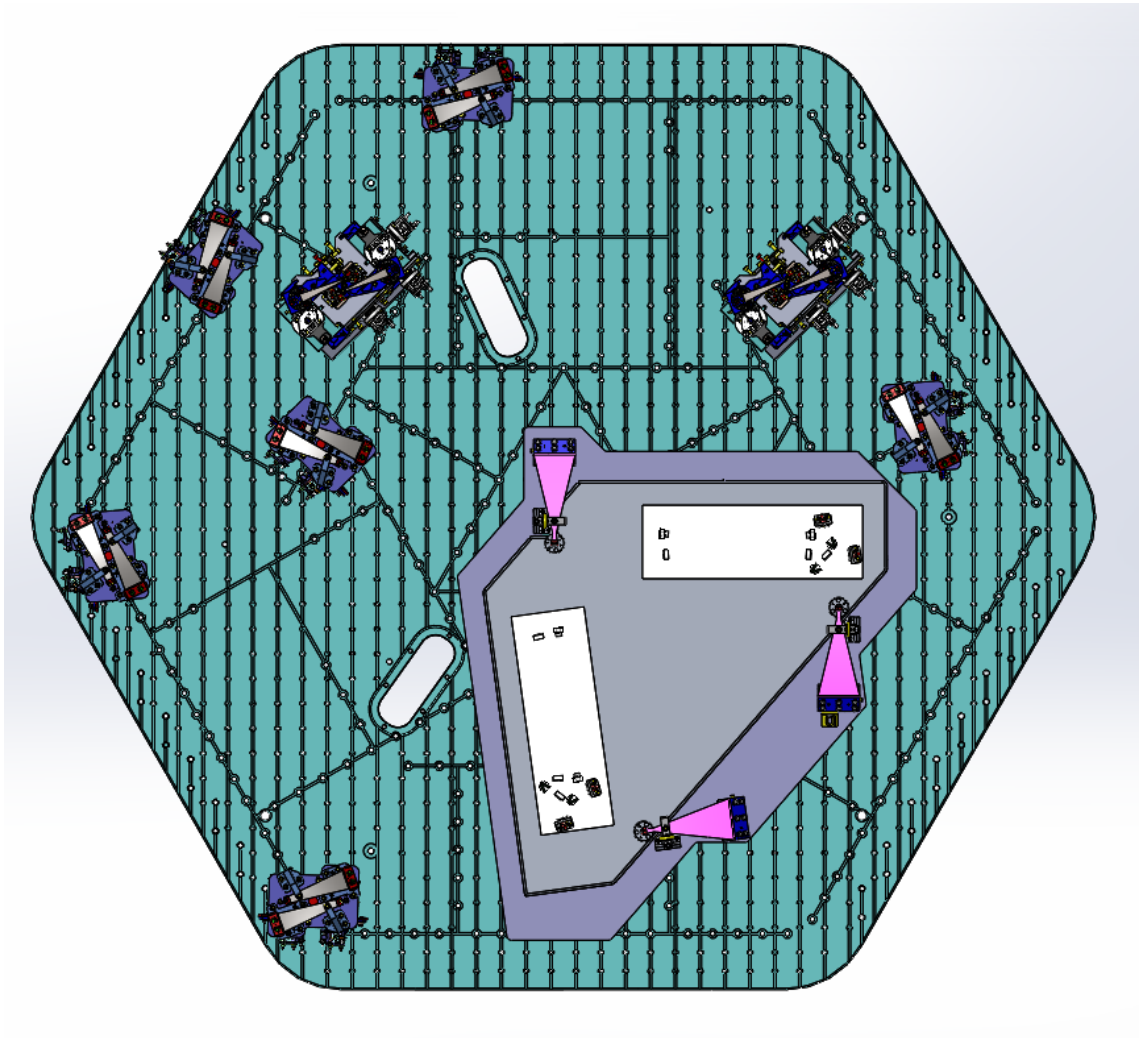


Figure 6.1: Place holder graphic. The pentagram layout. CAD with beam path will be inserted later.

this figure. The size of the beam at the final mode matching optic matches the size of the target mode as the size of the beam cannot be changed at a mirrors surface. The beam is always less than 1.2 mm within HAM6, an important fact to remember when considering how close structures can be to the beam path.

The astigmatism loss was calculated to be $> 0.05\%$; for detail in this type of calculation, see Appendix C. This astigmatism loss is plotted as a function of angle in Figure 6.4; above 10° the loss becomes significant. The compromise that is made when choosing a larger AOI is between the potential for larger Gouy phase separation between OMx1 and OMx2 at the expense of loss due to astigmatism. In reality the IFO and OMC will also have a small amount astigmatism so this calculation will not be entirely correct, however this effect is negligible compared to other losses. Losses of $> 0.05\%$ are considered unimportant, especially when compared to the other expected losses, some of which are expected to be at the many percent level, in the readout of the interferometer.

The focal lengths of the OFI lens, OMx1 and OMx2 where optimised so that there was as close to 45° Gouy phase separation between OMx1 and OMx2 as possible. This is shown in Figure 6.3. With the layout shown in Figure 6.1, it is possible to achieve 43° Gouy phase separation between OMx1 and OMx2. Other layouts could provide exactly 45° Gouy phase separation; however, the this layout offers many practical benefits, mainly in relation to the RPS, over the other considered designs that outweigh an additional 2° of Gouy phase separation between OMx1 and OMx2. This layout also provides good Gouy phase separation for alignment control, with there being 88° between OMxS and OMx2.

Optic	Power (1/m)	Focal length (m)	RoC (m)
OFI lens	-0.0583	-17.1552	-
OMx1	-0.3539	-2.8290	5.6580 (concave)
OMx2	-0.8737	-1.1445	2.2891 (concave)

Space Name	Optical Path Length (mm)
SRM to OFI lens	1120
OFI lens to HAM6 table edge	720
HAM6 table edge to OM0	432
OM0 to BHDBS2	905
BHDBS2 to OMxS	225
OMxS to OMx1	1240
OMx1 to OMx2	1590
OMx2 to OMCx waist	1220

Parameter	Value
SRM Complex Beam Parameter	$q_{\text{SRM}} = -3.55 + 1.09i \text{ m}$
OMC waist	$485 \mu\text{m}$
OMx1 to OMx2 Gouy phase	43°
OMxS to OMx2 Gouy phase	88°
OMx1 and OMx2 AOI	A: 10° , B: 6°

Table 6.2: Mechanical and optical properties of the LIGO A+ HAM6 layout.

In conclusion, the proposed HAM6 layout that fulfils the criteria set in Section 6.1 is shown in Figure 6.1. Near optimal mode-matching control and alignment control can be achieved with this layout. A summary of the properties of the layout can be found in Table 6.2. The next stage of development for this layout involves the design and prototyping of parts, and the consideration of fine details such as cable routing.

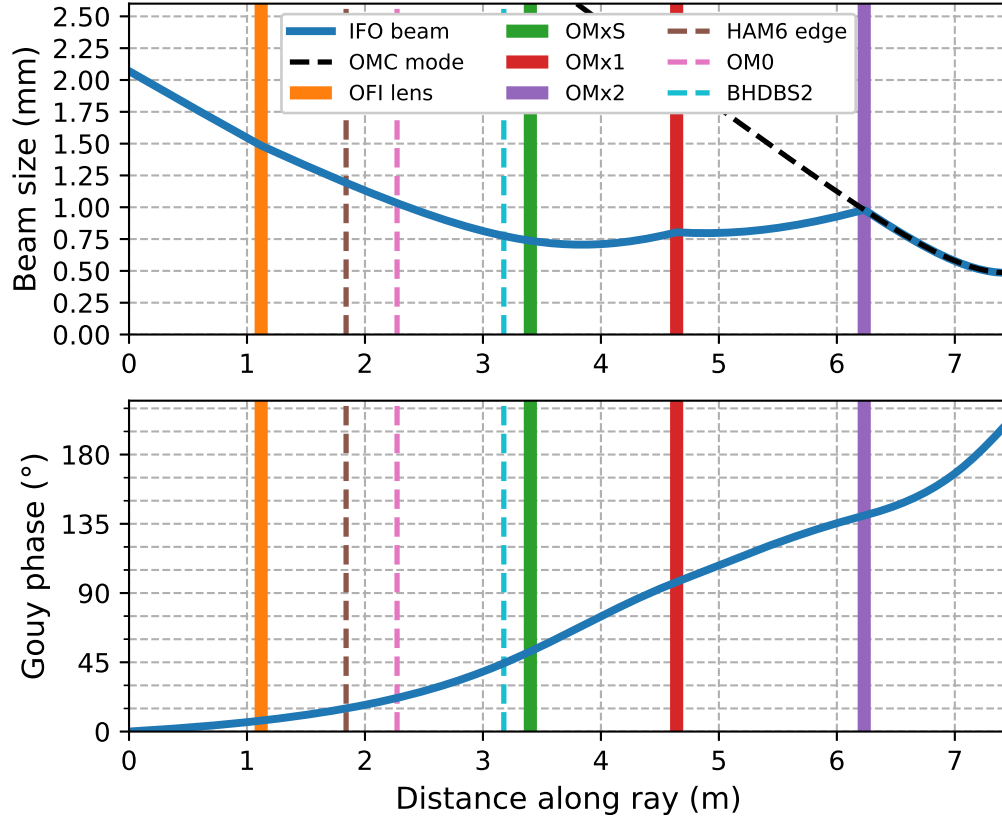


Figure 6.2: Top panel: the radius of the beam as it travels from the SRM ($z = 0$) to the waist of the OMC. Significant points along the axis are marked with vertical lines. Less significant, but useful reference points, are marked by dashed vertical lines. Bottom panel: The Gouy phase shift accumulated by the beam as it propagates along the optical axis.

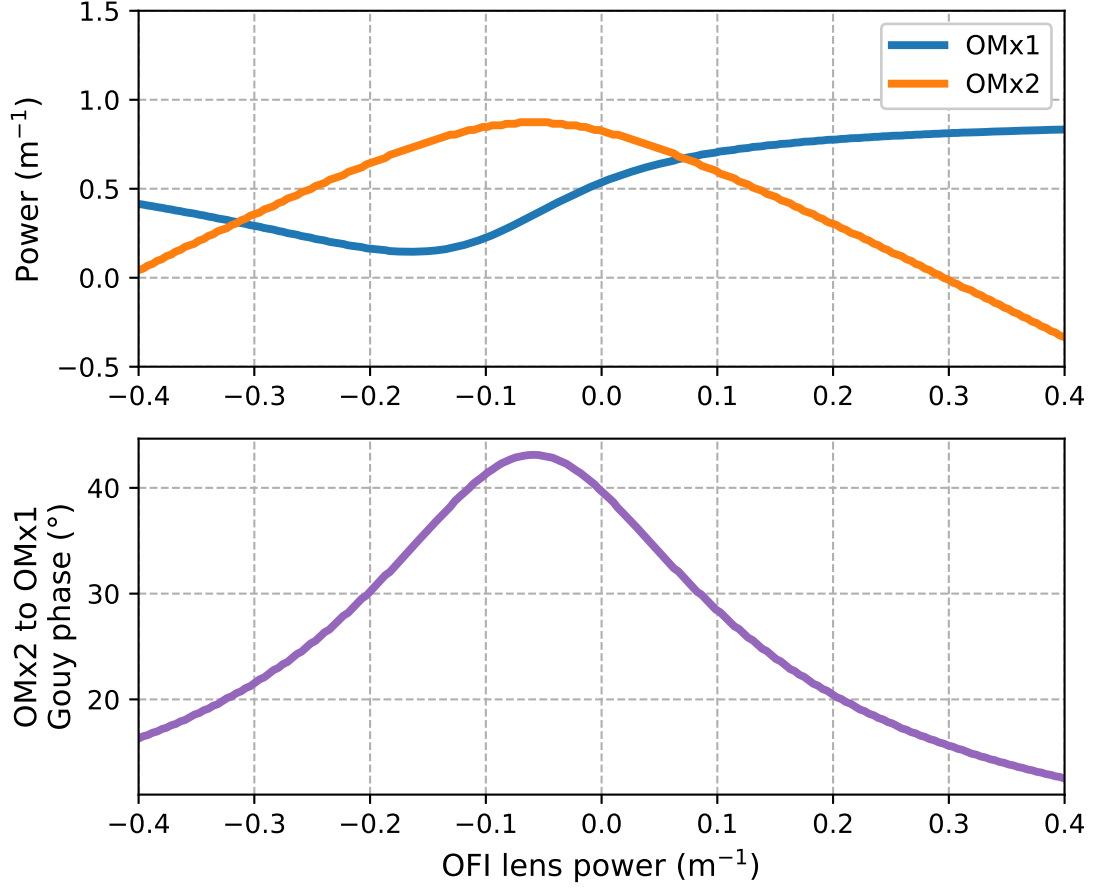


Figure 6.3: Top panel: The choice of OFI lens power affects the combination of OMx1 and OMx2 power required for 100% mode matching. Bottom panel: The best choice of OFI lens corresponds to where the Gouy phase separation between the pair of OMx1 and OMx2 mirrors that give 100% mode matching is closest to 45° . The optimal Gouy phase separation between OMx1 and OMx2 achievable by this layout is 43° .

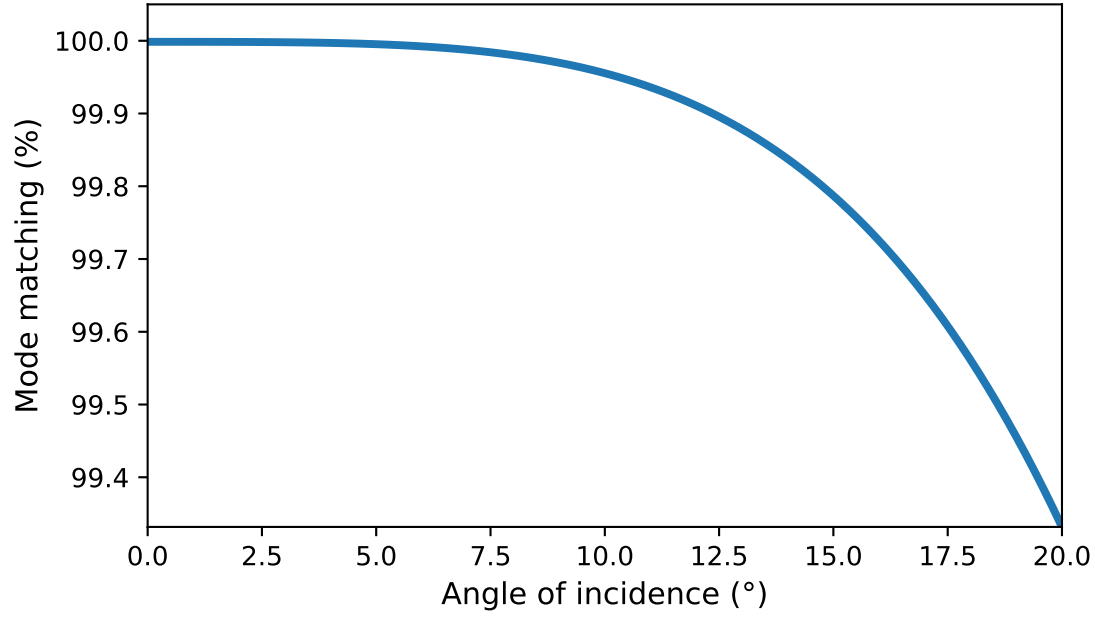


Figure 6.4: Astigmatism is a source of loss. When designing this layout, to explore how sensitive it was to astigmatism, the AOI for the curved mirrors was altered. Using equation (6.2), the loss was computed. This effect is negligible up to 10° after which it starts to become significant compared to the losses summarised in Table 6.1.

6.3 Visitations of the Robustness of the A+ HAM6 Layout to Mode Mismatch

It is expected that there will be some mismatch between the beam from the interferometer and the OMC due to geometry errors within the interferometer. To visualise the effectiveness of the mode matching control provided by the layout's active optics, it is useful to consider a 2D space representing Gaussian beams. The region of this space that can be mode matched by the layout's mode matching optics and a distribution of modes that those optics may have to correct for can be compared to gain insight into the effectiveness of the layout. It was found that the layout's active mode matching optics will cover the vast majority of expected input beams.

The mode matching between two laser beam modes, or a laser mode and a cavity mode, describes how similar those two modes are. In general, a combination of misalignment and differing beam shape can be considered to be a mode mismatch. However, mode matching typically refers to the overlap of the two modes if they were co-aligned, thus it is a measure of how similar in shape the two modes are.

6.3.1 A Description of Mode Matching in Terms of HG Modes

Mode-matching is usually, to first approximation, thought of as a difference in beam waist's size and beam waist's location. First order mode mismatch can be modelled as light scattering from the beam's, or cavity's, fundamental mode into its $\text{HG}_{20} + \text{HG}_{02}$ mode. If a cavity is designed such that the fundamental mode will be resonant while the $\text{HG}_{20} + \text{HG}_{02}$ mode will be reflected then the effect of mode mismatch between the cavity and a signal carrying beam is to decrease the signal light that is

transmitted by the cavity; since, some of the signal carrying light will be scattered into the reflected $\text{HG}_{20} + \text{HG}_{02}$ mode. Sensing the $\text{HG}_{20} + \text{HG}_{02}$ mode therefore gives a measure of the mode-matching between two lasers.

A change in waist size is equivalent to an in-phase coupling between the fundamental mode and the $\text{HG}_{20} + \text{HG}_{02}$, whereas a change in waist location is a coupling to the $\text{HG}_{20} + \text{HG}_{02}$ which is 90° out of phase with the fundamental mode [33]. As the $\text{HG}_{20} + \text{HG}_{02}$ acquires 2ϕ compared to the fundamental mode, where ϕ is the Gouy Phase, spacing beam curvature actuators at 45° will give orthogonal basis for control over mode matching. This means that the beam size and beam waist, or two combinations of them that are orthogonal, can be independently controlled.

6.3.2 Calculating Mode Matching Using The Gaussian Beam Parameter

Two perfectly mode matched beams will have identical beam parameters at the same point along the optical axis. The overlap integral between two fields, E_1 and E_2 ,

$$\mathcal{M} = \frac{|\int_A E_1^* E_2 dA|}{\int_A |E_1|^2 dA \int_A |E_2|^2 dA}, \quad (6.1)$$

can be used to determine the amount of mode matching, $\mathcal{M}$, between two astigmatic modes. Solving this integral gives

$$\mathcal{M} = \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}, \quad (6.2)$$

where $q_{1,2}$ is the beam parameter of the electric field $E_{1,2}$. The solution to equation (6.2) is always $0 < \mathcal{M} \leq 1$. Alternatively, mode matching may be computed in terms of the amplitude of the higher order modes relative to the fundamental mode [34]. The mode matching can be calculated at any point along the optical axis of the layout as mode matching must be preserved throughout it; however, it makes sense to calculate the mode matching in a physically significant plane such as the input plane, a plane containing a curved optic or the plane containing the waist of the target beam.

6.3.3 WS Space Visualisations

A Gaussian beam can be described by its beam size, w , and defocus, S . A WS space plots a Gaussian beam in terms of these two parameters. The effect of focussing elements on Gaussian beams can be clearly seen in this representation of them. Because a curved mirror can only change the curvature of a beam, at the mirrors surface the beams before and after reflection must have the same size. The process of a Gaussian beam, described by $q_1(S_1, w_1)$, being reflected from a mirror with power S_m to produce a new Gaussian beam, q_2 , can be written as

$$\frac{1}{q_2} = S_m + \frac{1}{q_1} = (S_m + S_1) - i \frac{\lambda}{\pi w_1^2}. \quad (6.3)$$

From equation (6.3) it can be seen that a mode moves in the S direction when it reflects from a curved mirror. This representation of Gaussian beams has the advantage of showing the beam in terms of intuitive quantities. The downside to this representation is as the beam propagates through the layout, the path that is traced through the WS space by the mode is fairly complex.

There are two other ways that a beam parameter could be graphically represented. Since q is a complex number, it could be represented in terms of its real and imaginary components. This makes seeing the effect of propagation straight forward, but the effect of a focussing element is harder to visualise. Alternatively, plotting $1/q$ in terms of $\lambda/\pi w^2$ and S produces plots where the effect of focussing elements and propagation are easy to visualise. Curved mirrors and lenses result in a change in defocus, and propagated modes follow circular paths. As these figures are being used to visualise mode matching and the effect of actuators on the beam size and defocus, the WS representation was chosen.

To illustrate how actuators move a mode in WS space, consider two lenses with variable power separated by 45° of Gouy phase. As the two lenses are orthogonal, the waist size and waist location of the mode exiting this layout will be orthogonally controllable. In a WS space the orthogonality of two actuators can be seen by the angle between the two vectors that represent each actuation. If the two lenses are orthogonal then the actuation vectors will be orthogonal. Figure 6.5 is a sketch of the WS space representation for a simple optical layout consisting of two active lenses separated by 45° Gouy phase.

The mode matching requirement between the interferometer and the OMC for LIGO A+ is $> 98\%$. It is useful to plot contours of the mode overlap in the WS space between a mode and a target mode. Note that these contours can only tell you about the overlap between a mode and the target mode; the mode matching between two arbitrary modes can not be deduced from these contours. For example two modes may be mismatched by the same amount from the target in orthogonal

directions in WS space. They cannot be 100% mode matched as these two modes are different from each other.

6.3.4 Visualising how Active Mode Matching Optics Expand the Area of WS Space Representing Acceptable Input Modes

To visualise the effectiveness of the actuation provided by the active elements of the HAM6 layout, the region of the WS space at the SRM from which the target mode matching space achieved was computed. To do this, first the ring of modes in WS space that is 98% mode matched to the OMC after reflection from the second active mirror, OMx2, was considered. If OMx2 had a static radius of curvature then the effect of OMx2 would be to move the ring vertically in the WS plane. However, the ring can also be stretched in the S direction as OMx2's radius of curvature can be changed. This can be seen, crudely, by applying $S_{\text{OMx2}} \pm \Delta S$ to the original $> 98\%$ mode matching ring to produce two new rings separated by $2\Delta S$.

To generate the region from which the target region can be reached by OMx2, the target region should be cut by a line which intersects the locations on the ring at the extrema of the beam size. To maximise the amount of WS space covered by the actuator, the upper target region should have $+\Delta S$ applied to it, whereas for the lower region the actuation should be $-\Delta S$. This results in region of WS space which is larger than the initial target region. All modes in the new region can be at least 98% mode matched to the OMC. The new region was then propagated to the first mirror, OMx1, where the same procedure can be applied to further expand the

region. This process is shown in Figure 6.6.

The modes from the SRM that can be matched to the target is shown in Figure 6.7. To visualise how successful the layout is at handling the likely modes from the interferometer (this set of modes was investigated in Chapter 7), the interferometer modes were compared to the region of WS phase space containing modes which can be corrected such that they meet the $> 98\%$ mode matching requirement. It was found that almost all likely modes will be covered by the layout, as the modes near the edge of the interferometer mode continuum are more unlikely to occur than the ones near the centre. Its worth noting that the in Figure 6.7, modes at the yellow end of the spectrum may not be physical as they correspond to an unstable SRC.

For all candidate layouts analysed there was no major difference in the amount of modes that could be mode matched. In principle, the requirement for having 45° of Gouy Phase separation between OMx1 and OMx2 does not maximise the amount of interferometer modes that can be mode matched. It may be possible to shape the region of WS space covered by the layout to maximise the amount of input modes that are covered by the layout. The shape of the pink region in Figure 6.7 is nominal; when HAM6 is constructed the pink regions shape will be altered by geometric errors in OMx1 and OMx2. Nevertheless, Figure 6.7 is encouraging since it shows that the actuation provided by OMx1 and OMx2 is highly tolerant to mode matching errors, so much so that the strict definition of the role of AWC in HAM6 may be relaxed in order for operators to use the AWC in HAM6 to correct for errors in the beam parameter at the SRM.

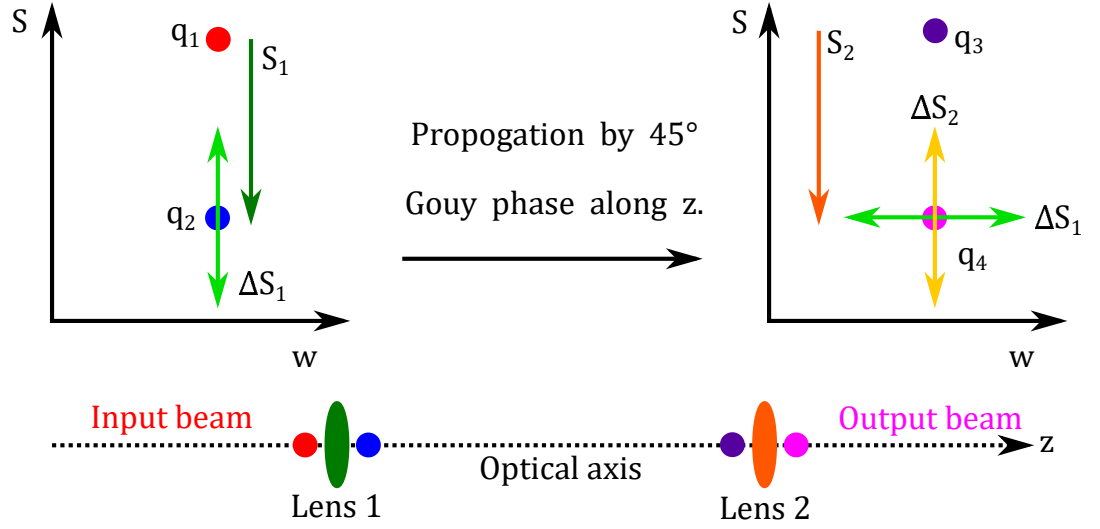


Figure 6.5: Sketches of the WS representation for a Gaussian beam interacting with different parts of a simple optical layout consisting of two active lenses with powers $S_{1,2} \pm \Delta S_{1,2}$. The left hand axis shows the WS plane at lens 1. The input beam q_1 is indicated by a red dot, and the effect of the lens on q_1 is shown by the dark green arrow S_1 . This results in a new mode q_2 which is shown as a blue dot. A light green arrow indicates the actuation of lens 1. The right hand axis shows the WS plane at lens 2. The input mode at lens 2, q_3 , is shown as a purple dot, and the output mode, q_4 , is shown as a pink dot. The static and dynamic power of lens 2 is indicated by the orange and yellow lines. As lens 1 and lens 2 are separated by 45° of Gouy Phase, the actuation provided by lens 1 results in a change in the beam parameter which is orthogonal to the actuation provide by lens 2, provided that the actuations are small enough. Beneath the two WS space diagrams is a sketch of the optical axis and where the modes are along it.

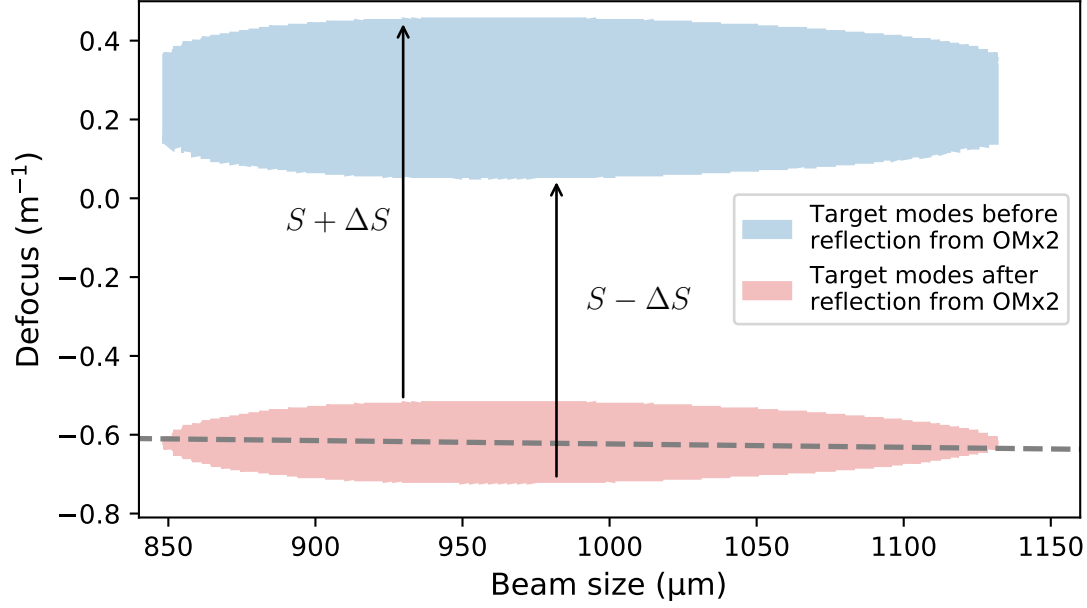


Figure 6.6: This is the WS space at the OMx2 plane. The effect of OMx2 in the plane of OMx2 is to move a mode vertically in WS space. After reflection from OMx2, a mode must be at least 98% mode matched to the OMC as there are no further active mode matching elements in the layout of HAM6. Modes with at least 98% are shown by the red region. The blue region represents which modes can be reached from the red region by either $S_{\text{OMx2}} \pm \Delta S$, and was constructed by applying transformations to the modes in the red region. The red region was split into two parts, ‘upper’ and ‘lower’, indicated by the dashed line that connects the two extremes in width of the region. To create the lower edge of the blue region, modes in the lower edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} - \Delta S$ applied to them. To construct the upper edge of the blue region, the upper edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} + \Delta S$ applied to them.

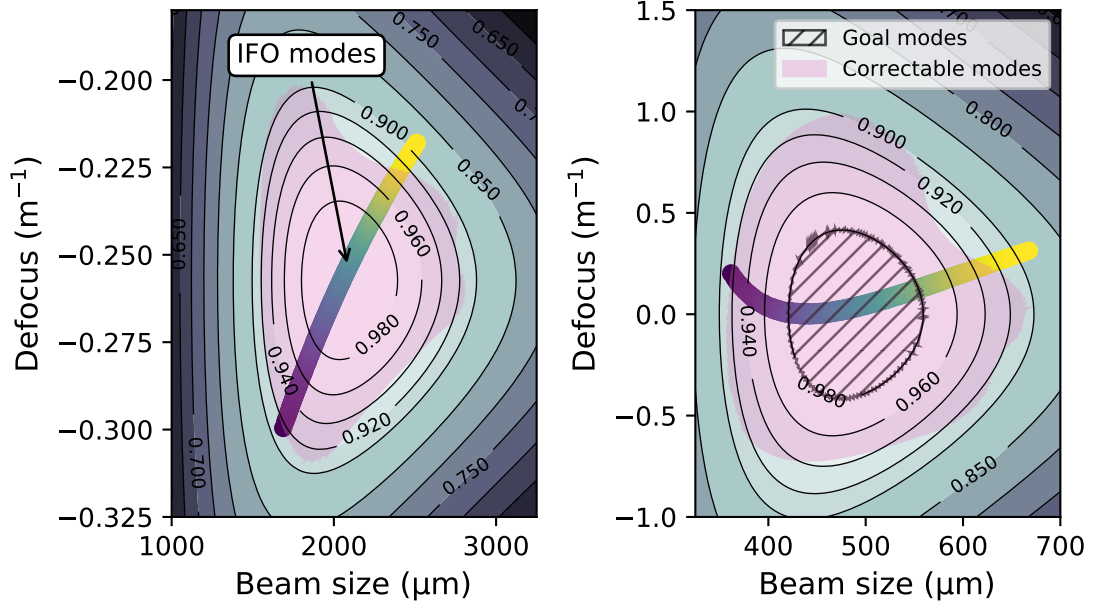


Figure 6.7: Left panel: the set of modes that could be present at the SRM is shown by the purple-blue-green-yellow curve, as described in Chapter 7. The pink region shows which modes can reach 98% mode matching by using the AWC in HAM6. The contours indicate the mode matching between a mode (W, S) and the OMC mode. Right panel: this shows the same modes as the top panel, except they have been transformed by the optical layout to the plane of the OMC waist. Modes within the pink region can be actuated on so they move to the goal region, which is indicated with a hatched area.

6.4 Conclusion

The functionality required from the BHD layout was discussed, and a proposed layout that satisfies these requirements is shown in Figure 6.1. The optical properties of this layout were investigated. By selection of the right combination of mode matching optics, near orthogonal control of the signal-LO mode matching and beam alignment to the OMC was achieved.

The effect that the AWC had on the region of WS space that contains modes which satisfy the $> 98\%$ mode matching requirement in WS space was investigated. Figure 6.7 shows that the AWC provides a huge amount of mode matching range. This means that the layout can tolerate large changes in the beam parameter at the SRM.

Chapter 7

The Consequence of Radius of
Curvature Errors within the SRC on
Modes at the SRM.

Chapter 8

In Air BHD experiments

Chapter 9

In-vacuum experiments

Chapter 10

Relevant Semiconductor and Photodiode Theory

Chapter 11

Characterisation of InGaAs Photodiodes

Chapter 12

Characterisation of 2um Photodiodes

N.B already in paper form.

Chapter 13

Characterisation of InSb Photodiode

Chapter 14

Conclusion

Appendix A

FINESSE Simulations

Appendix B

Algorithm For Finding a Three Optic Solution with 45° Gouy Phase

Separation between two of the Optics

To find a mode matching solutions between the SRM and the OMCs which have 45° phase separation for the AWC optics, an algorithm was developed. To visualise how this algorithm works see figures B.2 and B.1. Since a beam can be described by two parameters, its defocus S and width w , the modes can be shown in a 2-D space [35] (figure B.2). The action of a lens or curved mirror is to move the mode in the S direction. It is helpful to define a mode continuum along which w and/or S varies. The principle by which this algorithm works is: The continuum of modes after optic 1 and the continuum of modes just before optic 3 can be easily obtained since they must have a defined beam size, so they only vary in defocus. These two continuum are propagated to optic 2, and then its just a case of finding how to move from one continuum to an other.

The continua at optic 2 do not necessarily have to have the shape that they are sketched with in figure B.2, however if the blue-green and red-yellow continuum extend far enough they will always wrap around. It is also not guaranteed that every mode in the red-yellow can be reached from the blue-green continuum, and vice versa. Since the mapping between the blue-green and red-yellow continuum is not one to one, there is some choice in you can make. Both ‘branches’ of the continua, where the continuum turns back on itself in the w direction, do not need to be analysed at the same time, but a better solution may be found by looking at different combinations of branches. This is useful in setting a limit on the size of the continuum you search.

In our implementation of this algorithm, the continua are actually discrete, which means that when finding a mapping at optic two it is not guaranteed that a mode in the red-yellow continuum will have a corresponding mode of the same with in the blue-green continuum. Because of this granularity, if you were to plot optimal values of optic 2 and optic 3 power as a function of optic 1 power, you would see jumps in the optimal lens choices. This is especially bad around where the turning point in the continua is, since the modes defocus changes quickly while their size does not.

B.0.1 Method

This section delineates the method described in section B and can be visualised using figure B.2.

Define the target mode at the optic 3. In figure B.2, this is mode l at optic three.

Define a set of modes which share the same beam size as the target mode at optic 3 but have varying radii of curvature. In figure B.2, this is the red-yellow continuum at optic three.

Propagate the set of modes defined at optic 3 back to the optic 2. In figure B.2, this is the red-yellow continuum at optic 2.

Define the initial beam parameter and propagate this to optic 1. In figure B.2, this is mode i at optic one.

Define a set of modes that have the same beam size as the initial beam at optic 1 but differ in defocus. In figure B.2, this is the blue-green continuum at optic one.

Propagate the modes defined at the optic 1 to optic 2. In figure B.2, this is the blue-green continuum at optic two.

Calculate the difference in optical power between mode i and a mode in the blue-green continuum (e.g. mode j). In figure B.2, mode j represents just one possible mode in this continuum. Save this power.

Find a mode in the red-yellow continuum which is closest in size to mode j, call this mode k. In figure B.2, there is another possible choice on the red-yellow continua, however only one of them needs to be considered.

Calculate the difference in optical power between mode j and mode k at optic 2. Save this power.

Calculate the difference in optical power mode k and the target mode at optic 3. Save this power.

Calculate the Gouy phase separation of optic 2 and optic 3.

Repeat for every mode in the blue-green continuum, or until a gouy phase separation of 45degree has been found.

The number of modes that are considered and the range of modes in the continua should be tailored for the optical layout that is being considered. Our problem was with metre scale spacings between optics, and with metre scale Rayleigh lengths. It was found that continua with a range of the order 1-10 dioptries were suitable. This does not mean that an adaptive optic would need this actuation range, rather it allowed for the static radii of curvature for each optic that gives optimal Gouy phase separation between the active mode matching elements to be found. Using this algorithm, we were able to rapidly develop many possible HAM6 layouts. This allows for more time to consider various engineering factors (suspension size and the like) and how to adapt the mirror strengths accordingly to accommodate for small changes in a tentative design.

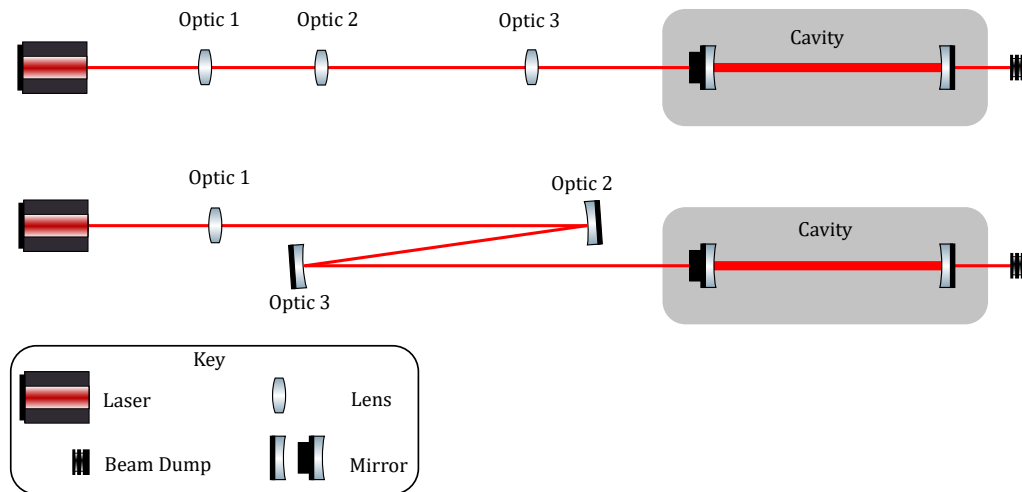


Figure B.1: Top layout: Three lenses used to achieve mode matching to the cavity. These are shown as positive lenses, however negative lenses can also be used when designing mode matching systems. Bottom Layout: A lens and two curved mirrors to achieve mode matching to a cavity. The optics 1,2 and 3 correspond to the ones shown in figure B.2. The initial and target modes are defined in these examples by the laser mode and the cavity mode.

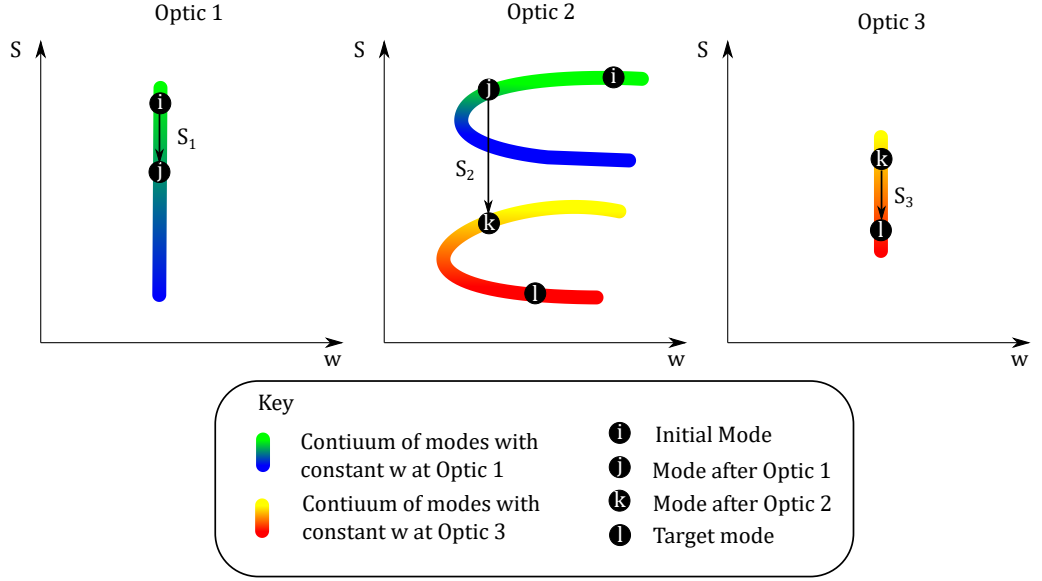


Figure B.2: Sketch of the WS phase space representations of a Gaussian beams at the three positions along the optical axis where a mirror or a lens could be. Examples of such layouts are shown in figure B.1. The optical power of the first optic is given by the difference in the defocus between the initial mode (i) and the second mode (j). The power of the second optic is found by finding the difference in defocus from the blue-green continuum to the red-yellow continuum. There are 2 optical powers that will move the mode j to the red-yellow continuum, and two possible modes in the blue-green continuum that can from which k can be reached, giving 4 possible similar solutions. The algorithm can be set up so that it is only one way of moving between the blue-green and red-yellow continua by limiting their extent. The power of the third optic is found from the difference between the mode k and mode l at optic 3.

Appendix C

Mode Matching Between 2 Astigmatic Beams

C.1 Derivation of Mode Matching TM00 Beams

The [check this naming](#) electric field, E , of a TM00 Gaussian beam can be written as

$$E(x, y, z) = E_x(x, z)E_y(y, z) \quad (\text{C.1})$$

where

$$E_x(x, z) = \frac{1}{\sqrt{q_x(z)}} \exp\left(-ik \frac{x^2}{2q_x(z)}\right), \quad (\text{C.2})$$

$$E_y(y, z) = \frac{1}{\sqrt{q_y(z)}} \exp\left(-ik \frac{y^2}{2q_y(z)}\right). \quad (\text{C.3})$$

For a beam without astigmatism, $x^2 + y^2 = r^2$ and $q_x = q_y = q$ so:

$$E(r, z) = \frac{1}{q(z)} \exp\left(-ik \frac{r^2}{2q(z)}\right). \quad (\text{C.4})$$

The mode matching, $\mathcal{M}$, of two beams can be expressed by overlap of the two fields

$$\mathcal{M} = \frac{\left| \int_A E_1^* E_2 dA \right|}{\int_A |E_1|^2 dA \int_A |E_2|^2 dA}. \quad (\text{C.5})$$

The numerator of equation C.5 for TM00 beams can be written as:

$$\left| \int_A E_1^* E_2 dA \right| = \int_A E_1^* E_2 dA \int_A E_1 E_2^* dA \quad (\text{C.6})$$

$$= \int_{-\infty}^{\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{\infty} E_{y1}^* E_{y2} dy \int_{-\infty}^{\infty} E_{x1} E_{x2}^* dx \int_{-\infty}^{\infty} E_{y1} E_{y2}^* dy. \quad (\text{C.7})$$

Since the calculation is the same for the x and y coordinates, we only need to show the solution for the x . The x terms of equation C.7 can be written as:

$$\int_{-\infty}^{\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{\infty} E_{x1} E_{x2}^* dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}^*}} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \frac{1}{\sqrt{q_{x2}}} \exp\left(\frac{-ikx^2}{2q_{x2}}\right) dx \quad (\text{C.8})$$

$$\times \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}}} \exp\left(\frac{-ikx^2}{2q_{x1}}\right) \frac{1}{\sqrt{q_{x2}^*}} \exp\left(\frac{ikx^2}{2q_{x2}^*}\right) dx \quad (\text{C.9})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \exp\left(\frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{C.10})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*} - \frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{C.11})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2 (q_{x2} - q_{x1}^*)}{2q_{x1}^* q_{x2}}\right) dx \times c.c. \right). \quad (\text{C.12})$$

Using the identity

$$\mathcal{I} = \int_{-\infty}^{\infty} \exp(iax^2) dx = \frac{i\pi}{a} \quad (\text{C.13})$$

and labelling

$$a = \frac{k(q_{x2} - q_{x1}^*)}{2q_{x1}^*q_{x2}} \quad (\text{C.14})$$

allows equation C.12 to be simplified to

$$\frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp \left(\frac{ikx^2 (q_{x2} - q_{x1}^*)}{2q_{x1}^*q_{x2}} \right) dx \times c.c. \right) = \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{i\pi (2q_{x1}^*q_{x2})}{k(q_{x2} - q_{x1}^*)}} \sqrt{\frac{-i\pi (2q_{x1}q_{x2}^*)}{k(q_{x2}^* - q_{x1})}} \quad (\text{C.15})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{4\pi |q_{x1}|^2 |q_{x2}|^2}{k^2 |q_{x2} - q_{x1}^*|^2}} \quad (\text{C.16})$$

$$= \frac{2\pi}{k} \frac{1}{|q_{x2} - q_{x1}^*|}. \quad (\text{C.17})$$

This means that the numerator of equation C.5 is

$$\frac{4\pi^2}{k^2} \frac{1}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{C.18})$$

Since E_x and E_y are orthogonal, $|E|^2 = |E_x|^2 |E_y|^2$, an expression for the denominator of equation C.5 can be found by:

$$\int_A |E|^2 dA = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |E_x|^2 |E_y|^2 dx dy \quad (\text{C.19})$$

$$= \int_{-\infty}^{\infty} \frac{1}{|q_x|} \exp \left(\frac{ikx^2 (q_x - q_x^*)}{2q_x q_x^*} \right) \times y \text{ term.} \quad (\text{C.20})$$

Using a similar approach as in the calculation of the numerator terms:

$$\int_A |E|^2 dA = \frac{1}{|q_x|} \sqrt{\frac{i2\pi |q_x|^2}{k(q_x - q_x^*)}} \frac{1}{|q_y|} \sqrt{\frac{i2\pi |q_y|^2}{k(q_y - q_y^*)}} \quad (\text{C.21})$$

$$= -\frac{2\pi}{k} \frac{1}{\sqrt{(q_x - q_x^*)(q_y - q_y^*)}} \quad (\text{C.22})$$

we get the mode matching $\mathcal{M}$ to be

$$\mathcal{M} = \frac{\frac{4\pi}{k^2}}{-\frac{2\pi}{k} \times -\frac{2\pi}{k}} \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)} \sqrt{(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|} \quad (\text{C.23})$$

$$= \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{C.24})$$

C.2 Effect of Error in the Radius of Curvature of a Mirror on the defocus of a Beam

The error in the optical power of a mirror due to an error in the radius of curvature of is

$$S = \frac{2}{R_m} \quad (\text{C.25})$$

$$dS = \frac{-2}{R_m^2} dR. \quad (\text{C.26})$$

The error in the beams defocus is the same as the error in the radius of curvature of the the mirror. The effect this has on the beam parameter q can be found by

defining a contour of modes distance dq from q i.e. $q + |dq| e^{i\theta}$, where dq is

$$q = \frac{q_i}{Sq_i + 1} \quad (\text{C.27})$$

$$dq = \frac{q_i^2}{(-Sq_i + 1)^2} dS. \quad (\text{C.28})$$

This gives you a family of modes which will contain the modes due to an error in the radius of curvature of the mirror, but since q is a complex number the contour you need to define to encapsulate the concept of an error in q also contains modes which are have different beam sizes as well as different defocuses. To do the calculation this way is cumbersome, it is much more straight forward to compute the effect of having a mirror ROC error by evaluating at every mirror

$$\begin{pmatrix} 1 & 0 \\ \pm dS & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -S & 1 \end{pmatrix} \begin{pmatrix} q \\ 1 \end{pmatrix} = \begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix} \quad (\text{C.29})$$

and then divide the top and bottom entries of the $\begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix}$ vector.

Appendix D

Sensing and Control in the HAM6 Layout

The sensors and actuators for controlling the various degrees of freedom that the HAM6 layout is responsible for is described in brief in the following sections.

D.0.1 Longitudinal Control

Longitudinal control refers to the sensing and actuation of lengths within the interferometer, i.e. position of optics along the optical axis. An example of longitudinal degree of freedom is the differential arm length. There are many longitudinal degrees of freedom within the interferometer and many of them are the lengths of cavities. The degrees of longitudinal degree of freedom that the HAM6 layout must provide sensing and control for are described in the next three sections.

D.0.1.1 Homodyne Angle

One method of controlling the homodyne angle is using the so called ‘RF 36’ signal. LIGO uses 9 MHz and 45 MHz modulations on the light to detect various lengths within the interferometer that need to be controlled. The beat between these two will produce a 36 MHz signal. The homodyne angle determines the amplitude of this signal, and so by demodulating a photodiode measuring the light in reflection from the OMC at 36 MHz the homodyne angle may be controlled. This is detailed in (Tengs Thesis ref). The actuation may be provided by OM0, BHDBS2 or by BHDM1. Alternatively, a dither signal at a few kHz on OM0 could provide a signal corresponding to the homodyne angle.

D.0.1.2 OMC Length

Keeping the OMC length locked to the combined LO-AS beam can be done by actuating on the length of PZT mirrors in the OMC. A few kHz signal may be created by the modulation on a HRTS in either the LO or AS path, one of the HDS after BHDBS2, or even a dither on the other PZT within the OMC. These modulations will be measurable in transmission of the OMC. Alternatively, a PDH signal may be created by using the 9 MHz signal may be measured in reflection from the OMC.

D.0.1.3 Squeezing Angle to Measurement Phase

The squeezing and measurement phase may be controlled by measuring the 3 MHz signal transmitted through the OMC. This signal originates in the AOM in the squeezer, and the squeezing angle can be controlled with the PZT in the squeezer.

layout in HAM7.

D.0.2 Alignment Control

The alignment of beams within the interferometer is required to maximise the power of the laser the mode containing the gravitational wave signal. The alignment degrees of freedom that need to be sensed and controlled by the HAM6 layout are discussed in the next four sections.

D.0.2.1 LO and AS alignment

The alignment of the signal to the to the AS beam will be done with either OM0 or BHDM1, and BHDBS2. A 36 MHz signal measured in reflection of the OMC can be used to determine the overlap of the two beams. By having a Gouy Phase telescope in reflection of the the OMC, a signal proportional to displacement and angle may be obtained. Alternatively, a dither at a few Hz can be applied to BHDM1 and BHDBS2, and this signal will appear in the light transmitted by the OMC.

D.0.2.2 Alignment of the LO-AS beam to the OMC

A dither applied to a HRTS in either the LO or AS path will produce a signal in transmission through the OMC. The mirrors OMxS and OMx2 will be used to align the LO beam into the OMC, and as these are separated by $\sim 90^\circ$ of Gouy phase, orthogonal control of the input beam alignment will be achievable.

D.0.2.3 Squeezer to AS beam

The pick-off at OM0 will be used to align the squeezer beam to the AS beam. The squeezer light has 42 MHz sidebands, and so a pair of wavefront sensors separated by $\sim 90^\circ$ of Gouy phase can be demodulated at this frequency to derive an alignment error signal. The squeezer beam will be aligned to the AS beam with optics between the filter cavity and the OFI. The signal and squeezer beam will be aligned onto these wavefront sensors using SR2 and a tip tilt in transmission of OM0.

D.0.2.4 Differential Hard Mode

The differential hard mode arises due to radiation pressure effects in the arm cavities. As the power in the cavity changes, the dynamics of the test mass changes. These dynamics result from the alignment of the cavity axis to the test masses due radiation pressure. This differential hard mode dynamic must be controlled. This signal will be detected with light that is transmitted through OM0, and actuated on with the test masses.

Appendix E

Quantum Optics

Bibliography

- [1] Haocun Yu, , L. McCuller, M. Tse, N. Kijbunchoo, L. Barsotti, and N. Mavalvala. Quantum correlations between light and the kilogram-mass mirrors of LIGO. *Nature*, 583(7814):43–47, jul 2020.
- [2] Stefan L. Danilishin and Farid Ya. Khalili. Quantum measurement theory in gravitational-wave detectors. *Living Reviews in Relativity*, 15(1), apr 2012.
- [3] E. Knyazev, S. Danilishin, S. Hild, and F.Ya. Khalili. Speedmeter scheme for gravitational-wave detectors based on EPR quantum entanglement. *Physics Letters A*, 382(33):2219–2225, aug 2018.
- [4] Tobias Eberle, Sebastian Steinlechner, Jöran Bauchrowitz, Vitus Händchen, Henning Vahlbruch, Moritz Mehmet, Helge Müller-Ebhardt, and Roman Schnabel. Quantum enhancement of the zero-area sagnac interferometer topology for gravitational wave detection. *Physical Review Letters*, 104(25), jun 2010.
- [5] H. J. Kimble, Yuri Levin, Andrey B. Matsko, Kip S. Thorne, and Sergey P. Vyatchanin. Conversion of conventional gravitational-wave interferometers into quantum nondemolition interferometers by modifying their input and/or output optics. *Physical Review D*, 65(2), dec 2001.

- [6] T. Zhang, E. Knyazev, S. Steinlechner, F. Y. Khalili, B. W. Barr, A. S. Bell, P. Dupej, J. Briggs, C. Gräf, J. Callaghan, J. S. Hennig, E. A. Houston, S. H. Huttner, S. S. Leavey, D. Pascucci, B. Sorazu, A. Spencer, J. Wright, K. A. Strain, S. Hild, and S. L. Danilishin. Quantum noise cancellation in asymmetric speed metres with balanced homodyne readout. *New journal of physics*, 20(10):103040, 2018.
- [7] M S Stefszky, C M Mow-Lowry, S S Y Chua, D A Shaddock, B C Buchler, H Vahlbruch, A Khalaidovski, R Schnabel, P K Lam, and D E McClelland. Balanced homodyne detection of optical quantum states at audio-band frequencies and below. *Classical and Quantum Gravity*, 29(14):145015, jun 2012.
- [8] Daniel Sigg and. Status of the LIGO detectors. *Classical and Quantum Gravity*, 25(11):114041, may 2008.
- [9] Brian J. Meers and Kenneth A. Strain. Modulation, signal, and quantum noise in interferometers. *Physical Review A*, 44(7):4693–4703, oct 1991.
- [10] Sebastian Steinlechner, Bryan W. Barr, Angus S. Bell, Stefan L. Danilishin, Andreas Gläfke, Christian Gräf, Jan-Simon Hennig, E. Alasdair Houston, Sabina H. Huttner, Sean S. Leavey, Daniela Pascucci, Borja Sorazu, Andrew Spencer, Kenneth A. Strain, Jennifer Wright, and Stefan Hild. Local-oscillator noise coupling in balanced homodyne readout for advanced gravitational wave detectors. *Physical Review D*, 92(7), oct 2015.
- [11] D. F. Walls. Squeezed states of light. *Nature*, 306(5939):141–146, nov 1983.

- [12] Carlton M. Caves and Bonny L. Schumaker. New formalism for two-photon quantum optics. i. quadrature phases and squeezed states. *Physical Review A*, 31(5):3068–3092, may 1985.
- [13] Peter Fritschel, Matthew Evans, and Valery Frolov. Balanced homodyne readout for quantum limited gravitational wave detectors. *Optics Express*, 22(4):4224, feb 2014.
- [14] Tobin T Fricke, Nicolás D Smith-Lefebvre, Richard Abbott, Rana Adhikari, Katherine L Dooley, Matthew Evans, Peter Fritschel, Valery V Frolov, Keita Kawabe, Jeffrey S Kissel, Bram J J Slagmolen, and Sam J Waldman. DC readout experiment in enhanced LIGO. *Classical and Quantum Gravity*, 29(6):065005, feb 2012.
- [15] Kiwamu Izumi and Daniel Sigg. Advanced LIGO: length sensing and control in a dual recycled interferometric gravitational wave antenna. *Classical and Quantum Gravity*, 34(1):015001, dec 2016.
- [16] S. Dwyer. O3b noise budget. <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=55755>.
- [17] S. Dwyer. personal communication.
- [18] B. Lantz. Optical Integration of the aLIGO isolation tables. LIGO Document G2001539-v2 <https://dcc.ligo.org/G2001539>.
- [19] D.E. McClelland, N. Mavalvala, Y. Chen, and R. Schnabel. Advanced interferometry, quantum optics and optomechanics in gravitational wave detectors. *Laser & Photonics Reviews*, 5(5):677–696, mar 2011.

- [20] The LIGO Scientific Collaboration. A gravitational wave observatory operating beyond the quantum shot-noise limit. *Nature Physics*, 7(12):962–965, sep 2011.
- [21] L. McCuller, C. Whittle, D. Ganapathy, K. Komori, M. Tse, A. Fernandez-Galiana, L. Barsotti, P. Fritschel, M. MacInnis, F. Matichard, K. Mason, N. Mavalvala, R. Mittleman, Haocun Yu, M.E. Zucker, and M. Evans. Frequency-dependent squeezing for advanced LIGO. *Physical Review Letters*, 124(17), apr 2020.
- [22] T. Zhang. *Advanced techniques for future generation gravitational wave detectors*. PhD thesis, University of Glasgow, 2019.
- [23] Hang Yu Peter Fritschel, Vivishek Sudhir. LIGO DCC Document G1702485-v1.
- [24] P. Fritschel and S. Hild. Requirements for the HAM Relay Triple Suspension (HRTS). LIGO Document T1800413-v1 <https://dcc.ligo.org/LIGO-T1900817>.
- [25] L. Barsotti, L. McCuller, M. Evans, and P. Fritschel. The A+ design curve. LIGO DCC, April 2018. LIGO Document T1800042-v5 <https://dcc.ligo.org/LIGO-T1800042/public>.
- [26] C. Bond, P. Fulda, A. Freise, D. Brown, K. Kokeyama, L. Carbone, and A. Perreca. Finesse Input Files for the L1 Interferometer. LIGO Document T1300901-v5 <https://dcc.ligo.org/LIGO-T1300901>.
- [27] B. Lantz, R. Mittleman, J. Kissel, J. Warner, and A. Pele. Typical Motion of HAM Suspension Points in O2. LIGO Document T1800066-v2 <https://dcc.ligo.org/T1800066>.

- [28] A. Huddart. A+ HRTS Preliminary Design Document. LIGO DCC, February 2019. LIGO Document T1900817-v5 <https://dcc.ligo.org/LIGO-T1900817>.
- [29] L. Barsotti. aLIGO ISC Anti-symmetric Port: Beam Parameters and Optical Layout. LIGO Document T1200410-v2 <https://dcc.ligo.org/LIGO-T1200410/public>.
- [30] A. Brookes, L. Barsotti, G. Billingsley, K.A. Strain, and M. Zucker. A+ o5-ham6 design considerations and requirements. LIGO DCC, April 2020. E1900385-v1.
- [31] Evgenia Kochkina. *Stigmatic and Astigmatic Gaussian Beams in Fundamental Mode: Impact of Beam Model Choice on Interferometric Pathlength Signal Estimates*. PhD thesis, Hannover U., 2013.
- [32] P. Fulda, R. Goetz, A. Green, R. Martin, and D. B. Tanner. A+ Output Faraday Isolator Final Design. LIGO Document T1900788-v2 <https://dcc.ligo.org/LIGO-T1900788>.
- [33] Euan Morrison, Brian J. Meers, David I. Robertson, and Henry Ward. Automatic alignment of optical interferometers. *Applied Optics*, 33(22):5041, aug 1994.
- [34] F. Bayer-Helms. Coupling coefficients of an incident wave and the modes of a spherical optical resonator in the case of mismatching and misalignment. *Applied Optics*, 23(9):1369, may 1984.
- [35] Antonio Perreca, Aidan F. Brooks, Jonathan W. Richardson, Daniel Töyrä, and Rory Smith. Analysis and visualization of the output mode-matching require-

ments for squeezing in advanced LIGO and future gravitational wave detectors.

Physical Review D, 101(10), may 2020.